

Minimal residual certificates for the Erdős–Straus conjecture: a verified census to 10^{12}

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Abstract

The Erdős–Straus conjecture asserts that $4/n = 1/a + 1/b + 1/c$ is solvable in positive integers for every $n \geq 2$; the open cases concentrate on primes in six residue classes mod 840 (*hard primes*). Every solution at a hard prime p has first denominator $a = (p + R)/4$ for an *admissible residual* $R \equiv 3 \pmod{4}$, and solvability at that R is the divisor-class condition $\exists k \mid m^2$, $m = pa$, with $k \equiv -m \pmod{R}$ — a criterion due to Bradford. The conjecture at p is therefore exactly the finiteness of the *minimal working residual* $R_{\min}(p)$, a structural invariant that verification records do not report.

We compute and independently verify $R_{\min}$ for all 1,175,215,396 hard primes below 10^{12} . The maximum is $R_{\min} = 111$, first attained at $p = 119,945,383,009$; it had stood at 107 across the preceding five orders of magnitude, and a heuristic model calibrated at 10^9 predicted in advance both the filling of a then-empty band $87 \leq R \leq 103$ and the decade in which the record would move. Verification is graded: exhaustive exact re-checking below 10^9 , complete deep tails and large systematic samples above, every certificate re-verified as $4abc = p(bc + ac + ab)$ in exact integer arithmetic, and all datasets published with checksums.

Around the census we assemble the apparatus it exercises. There are exact solvability criteria at $R = 3, 7, 11$ and at the first composite residual $R = 15$; those at $R = 3, 7$ are Yamamoto’s, restated in residual form, while the exact criteria at $R = 11$ and $R = 15$ are new. A support theorem — the critical-number theorem of Diderrich–Mann, transported to this setting — shows that failure at R forbids at least half the unit classes mod R on the shifted form $(p + R)/4$, uniformly in R , and its classical exceptions fall exactly at the residuals where an exact criterion is available. The uniform consequence bounds $R_{\min}$ itself: all but $O(x \exp(-c(\log \log x)^2))$ hard primes $p \leq x$ satisfy $R_{\min}(p) \leq \varepsilon \log \log x$, with explicit certificates — a threshold exponentially smaller than the one implicit in Vaughan’s covering congruences at comparable exceptional-set quality. The elementary layer and the finite enumerations behind the criteria are machine-checked in Lean 4/mathlib.

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1 Introduction

Erdős and Straus conjectured (see [5]) that for every integer $n \geq 2$ the equation

$$\frac{4}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \tag{1.1}$$

has a solution in positive integers a, b, c . By multiplicativity it suffices to treat prime n , and classical polynomial identities settle every prime outside the six residue classes

$$\mathcal{H} \bmod 840 = \{1, 121, 169, 289, 361, 529\},$$

the squares of units modulo 840 (Mordell [23, Ch. 30]; the quadratic-residue nature of the obstruction goes back to Yamamoto [32]). We call primes in these classes *hard primes*. Direct verification has confirmed the conjecture far beyond any range considered here: to 10^{14} by Swett [27], to 10^{17} by Salez [24], and to 10^{18} by Mihnea and Bogdan [22]; the present paper is logically independent of those bounds. The structure of the solution count was analyzed by Elsholtz and Tao [4]. For calibration on the density axis: the first density bounds on the exceptional set are due to Webb [31], sieve bounds were advanced by Sander [26], and Vaughan’s classical mean-value bound [30] gives an exceptional set $\ll x \exp(-c(\log x)^{2/3})$ for (1.1) — asymptotically stronger than any fixed power of $\log x$, and still the record on that axis.

This paper is about a different quantity, and it is primarily a computation. A verification to 10^{18} answers “is (1.1) solvable at n ?” and records a yes; it does not record *how hard* the instance was, and the published records do not report the parameter in which the difficulty is measured. By Proposition 1.1 — Bradford’s criterion [9], restated — every solution at a hard prime p is a certificate at some admissible residual $R = 4a - p \equiv 3 \pmod{4}$, and the conjecture at p is exactly the finiteness of the least such R , written $R_{\min}(p)$. Our main object is a census of $R_{\min}$: its value for every one of the 1,175,215,396 hard primes below 10^{12} , verified, archived, and reproducible. What comes out of it is a distribution with sharp structure (Section 4), a broken record (the maximum moves from 107 to 111 in the last decade computed), and a heuristic model calibrated at 10^9 whose advance predictions about the tail were borne out at 10^{11} and 10^{12} .

The rest of the paper is the apparatus the census exercises and the structure it exhibits: exact per-residual solvability criteria (Section 3), which are what make the head of the distribution computable in closed form; a support theorem valid at every residual, which converts failure into a sieve condition; and, as its capstone, an almost-all bound on $R_{\min}$ itself (Theorem 5.5). Section 5 states these and sketches the sieve consequences; complete proofs of the elementary layer, together with the finite enumerations behind the criteria, are formalized in Lean 4 and inventoried in Appendix A.

New contributions

The census. $R_{\min}(p)$ computed and independently verified for every one of the 1,175,215,396 hard primes $p < 10^{12}$ (Section 4): the full distribution (Table 3); a graded verification protocol (Section 4.1) in which every certificate is re-checked as $4abc = p(bc + ac + ab)$ in exact integer arithmetic; and archived datasets carrying checksums. The maximum of $R_{\min}$ moves for the first time in five orders of magnitude — it is 107 throughout $10^7 \leq x \leq 10^{11}$ and 111 at 10^{12} , first attained at $p = 119,945,383,009$ — and a calibrated independence model, fitted to the complete 10^9 solvability masks, predicted in advance both the population of the then-empty

band $87 \leq R_{\min} \leq 103$ and the decade in which the record would move. Both predictions were tested by the 10^{11} and 10^{12} runs and held. Underneath sits a reproducibility layer: a Lean 4 [3] / mathlib [21] formalization of the elementary layer and of the finite enumerations behind the exact criteria, and a published run kit (Appendix A).

The apparatus, and what in it is new. *Exact* solvability criteria at $R = 11$ (Theorem 1.5) and at the first composite residual $R = 15$ (Theorem 1.6: a Jacobi-character dichotomy with no budget cases), together with the finite-reduction lemma (Theorem 1.3) that licenses their enumeration. The criteria at $R = 3$ and $R = 7$ (Theorems 1.2, 1.4) are *not* new — they are Yamamoto’s, in a different notation; Remark 1.7 gives the dictionary and delimits what is new at $R = 11$ and $R = 15$. A reciprocity identity (Theorem 1.9) collapses the per-residual conditions into one global statement about p , which we use to describe the record prime mechanically. The support theorem (Theorem 1.10) shows that failure at R confines the factor classes of $(p+R)/4$ to at most half the unit classes mod R , for every admissible R at once; as an additive statement this is the critical-number theorem of Diderrich–Mann [13], and the contribution here is the transport to residuals (bounded exponent budgets, a single prescribed target) together with the observation that its classical exceptional groups occur exactly at $R = 3, 7, 15$, where an exact criterion is available — so the sieve dimension $\frac{1}{2}$ holds at every admissible residual with no gaps. From these follow certificate-producing exceptional-set bounds for every finite admissible list (Theorem 1.11, Proposition 1.12, Theorem 1.13) and, as the capstone, the *uniform chain* (Theorem 5.5): all but $O(xe^{-c(\log \log x)^2})$ hard primes $p \leq x$ have $R_{\min}(p) \leq \varepsilon \log \log x$, with an explicit certificate for each — Remark 1.14 states precisely what this does and does not improve on. Carved out of the ceiling of that argument is one open problem, Conjecture 5.7, verified exhaustively in the accessible range, whose resolution would raise the bound to $R_{\min}(p) \leq (\log x)^{c_0}$ (Theorem 5.8).

1.1 The residual method

Fix a hard prime p . All six classes satisfy

$$(H1) \ p \equiv 1 \pmod{8}, \quad (H2) \ p \equiv 1 \pmod{3}, \quad (H3) \ p \bmod 7 \in \{1, 2, 4\}.$$

For an *admissible residual* $R \equiv 3 \pmod{4}$ put

$$a = \frac{p+R}{4} \in \mathbb{Z}, \quad m = pa.$$

Substituting a into (1.1) and clearing denominators shows that a solution with first denominator a exists if and only if

$$\exists k \mid m^2 \quad \text{with} \quad k \equiv -m \pmod{R}, \tag{1.2}$$

in which case $b = (k+m)/R$ and $c = (m^2/k+m)/R$ are positive integers solving (1.1). We write $R_{\min}(p)$ for the least admissible R for which (1.2) holds. The formulation loses no solutions:

Proposition 1.1 (completeness). *Every solution of (1.1) at a hard prime p , ordered $a \leq b \leq c$, has $p/4 < a \leq 3p/4$; hence $R = 4a - p$ is an admissible residual in $(0, 2p]$ and the solution is a residual certificate at R . Consequently $R_{\min}(p) \leq 2p$ if and only if the conjecture holds at p .*

Proof. With $a \leq b \leq c$, the equation gives $1/a < 4/p \leq 3/a$, so $p/4 < a \leq 3p/4$ and $R = 4a - p \in (0, 2p]$; $R \equiv 3 \pmod{4}$ is automatic from $p \equiv 1 \pmod{4}$. By the displayed equivalence, the solution yields a divisor k as in (1.2), so $R_{\min}(p) \leq R \leq 2p$. Conversely (1.2) at any admissible R produces a solution. $\square$

A word on provenance. Proposition 1.1 is not new. The bounded search space together with the divisor-congruence criterion modulo $4a - p$ is due to Bradford [9], who proves that a solution exists at p if and only if some a in $\lceil p/4 \rceil \leq a \leq \lceil p/2 \rceil$ admits a divisor of a^2 in a prescribed class modulo $4a - p$ — his range is in fact slightly sharper than the $a \leq 3p/4$ recorded above, and the correspondence with solutions is likewise bijective. These are by now called the Bradford conditions [22]. Criteria of this general shape go back further. Yamamoto [32] constructs a complete system Σ of solvable coverings and proves that $4/p$ is solvable if and only if p lies in their union; his simplified subsystem Σ_0 is defined on precisely the six hard classes, and the individual coverings $\{-s/q\} = \{n : q \mid n + s\}$ are, for odd q , exactly the condition “ q divides $(p + s)/4$ ”. So Yamamoto’s system is a residual-indexed criterion in different notation, and several of the per-residual statements below are his; Remark 1.7 makes the dictionary explicit and says which are. Lopez [15] states the same shift-indexed divisor condition in general form for one solution shape. Mordell’s Chapter 30 [23, Ch. 30] collects the classical identities and his own non-existence theorem; the systematic divisor-counting form is Elsholtz–Tao’s [4]; and Salez [24] proves a completeness theorem for seven modular equations at the level of degree-one prime polynomials.

What this paper does with the criterion is to *measure* it. The organizing move is to index by the residual R rather than by the denominator a , which makes $R_{\min}(p)$ a well-defined statistic of p and the object of a census: it is computable at scale because each fixed R admits a finite exact criterion (Theorem 1.3), determined in closed form at $R = 3, 7, 11, 15$, and because failure at every other R is constrained uniformly (Theorem 1.10). Bradford’s range gives $R_{\min}(p) \ll p$; the computation of Section 4 shows $R_{\min} \leq 111$ throughout $p < 10^{12}$, and Theorem 5.5 bounds it by $\varepsilon \log \log x$ for almost every hard prime.

Two elementary congruences drive everything below. From $4a = p + R$,

$$a \equiv 4^{-1}p, \quad m \equiv 4^{-1}p^2 \pmod{R}, \quad (1.3)$$

and since $\gcd(p, R) = 1$ (as p is prime and $R \neq p$, $R \equiv 3 \not\equiv p \pmod{4}$), also $\gcd(a, R) = \gcd(m, R) = 1$: the setting is always a unit computation modulo R . By (1.3), m is a *square* modulo every prime $r \mid R$; hence for every prime $r \mid R$ with $r \equiv 3 \pmod{4}$ (at least one exists, since $R \equiv 3 \pmod{4}$),

$$\left(\frac{-m}{r}\right) = \left(\frac{-1}{r}\right) = -1 \quad (1.4)$$

— *the target class $-m$ is a quadratic non-residue mod r .*

A worked example. Take the hard prime $p = 1009 \pmod{840}$ and the smallest admissible residual $R = 3$. Then $a = (1009 + 3)/4 = 253 = 11 \cdot 23$ and $m = pa = 255,277$. Since $11 \equiv 2 \pmod{3}$, Theorem 1.2 below promises a certificate: take $k = 11$, a divisor of m^2 with $k \equiv -m \equiv 2 \pmod{3}$. Then $b = (k + m)/3 = 85,096$ and $c = (m^2/k + m)/3 = 1,974,822,872$, and indeed

$$\frac{4}{1009} = \frac{1}{253} + \frac{1}{85,096} + \frac{1}{1,974,822,872}.$$

Had every prime factor of a been $\equiv 1 \pmod{3}$, every divisor of m^2 would be $\equiv 1 \pmod{3}$ and the required class 2 would be unreachable — the prototype of all the obstructions studied below.

1.2 Main results

Theorem 1.2 ($R = 3$; exact). *Let $p \equiv 1 \pmod{4}$ be prime with $3 \nmid p$, and $a = (p + 3)/4$. Then (1.2) holds for $R = 3$ if and only if a has a prime factor $q \equiv 2 \pmod{3}$. When it does, $k = q$ is*

a certificate.

Modulo 3, character equals class and the criterion is elementary. For every other fixed residual we need to know that the search space is finite before we can enumerate it, which is the content of the next statement. We label it a theorem for reference, but it is bookkeeping on top of Bradford's criterion, and finite-generation statements of this kind have a long history in the subject: Terzi [28] generated 198 solvable residue classes modulo 120120 algorithmically, Yamamoto's [32, §3] whole apparatus is a finite generation procedure for solvable coverings, and Salez [24] proves a completeness theorem for polynomial families. What is specific here is the finite datum: the capped multiset of factor classes of $(p + R)/4$ together with $p \bmod R$.

Theorem 1.3 (finite reduction). *For every fixed prime $R \equiv 3 \pmod{4}$, the validity of (1.2) at a hard prime p depends only on (i) the multiset of classes mod R of the prime factors of $a = (p + R)/4$, with multiplicities capped at $\lceil (R - 1)/2 \rceil$, and (ii) $p \bmod R$. Consequently an exact solvability criterion exists for every R and is computable by finite enumeration.*

Theorem 1.4 ($R = 7$, hard primes; exact). *Let p be a hard prime and $a = (p + 7)/4$. Then (1.2) holds for $R = 7$ if and only if a has a prime factor that is a quadratic non-residue modulo 7.*

Theorem 1.5 ($R = 11$, hard primes; exact). *Let p be a hard prime and $a = (p + 11)/4$; note $3 \mid a$, $2 \nmid a$, $7 \nmid a$ automatically. Then (1.2) fails for $R = 11$ if and only if either*

- (a) *every prime factor of a is a quadratic residue mod 11; or*
- (b) *$v_3(a) = 1$, every other prime factor of a is $\equiv 1 \pmod{11}$ except a non-residue part — its factors in the non-residue classes mod 11 — of exactly one of the shapes*

$$\{q \equiv 2\}, \quad \{q \equiv 6\}, \quad \{q \equiv 2\} \cup \{q' \equiv 6\} \pmod{11},$$

each with multiplicity one, paired respectively with $p \equiv 2, 6, 1 \pmod{11}$.

For composite R the certificate condition couples the prime-power components of R through the Chinese Remainder Theorem, and the divisor-class computation lives in the (generally non-cyclic) unit group $(\mathbb{Z}/R)^*$. Theorem 1.3 holds verbatim with $(\mathbb{Z}/R)^*$ in place of the cyclic group (multiplicities capped once exponent ranges cover each class's cyclic span; see Remark 3.1). The first composite case turns out to be *simpler* than $R = 11$: a pure character dichotomy, with the quadratic character modulo a prime replaced by the Jacobi character modulo 15, and no budget cases at all.

Theorem 1.6 ($R = 15$, hard primes; exact). *Let p be a hard prime and $a = (p + 15)/4$; note $p \bmod 15 \in \{1, 4\}$ and $2 \mid a$ automatically. Then (1.2) holds for $R = 15$ if and only if a has a prime factor q with $\left(\frac{q}{15}\right) = -1$, i.e. $q \equiv 7, 11, 13, 14 \pmod{15}$.*

Remark 1.7 (attribution at $R = 3, 7, 11, 15$). Theorems 1.2 and 1.4 are not new. In Yamamoto's notation [32, §3] the covering $\{-s/q\}$ — the set of n with $q \mid n + s$, equivalently, for odd q , $q \mid (n + s)/4$ — lies in his system Σ_1 of solvable coverings exactly for $q \equiv -1 \pmod{3}$ when $s = 3$, and for $q \equiv -1, -2, -4 \pmod{7}$ when $s = 7$; he records the common form explicitly, noting that “all the coverings $\{-s/q\}$ with the Jacobi symbol $\left(\frac{-s}{q}\right) = -1$ are contained in Σ_1 , for $s = 1, 2, 3, 4, 7$ and 8 ” [32, p. 45], and Σ_1 is defined on precisely the six hard classes. Since $\left(\frac{-s}{q}\right) = \left(\frac{q}{s}\right)$ for $s \equiv 3 \pmod{4}$, and since membership of a non-residue class is inherited by some

prime factor, this is the sufficiency half of Theorems 1.2 and 1.4 verbatim; the necessity half is the character obstruction (Proposition 2.1), which is Yamamoto’s Lemma 4 specialised. What the statements below add at these two residuals is only that the certificate can be taken at residual exactly R , and a machine-checked proof.

At $R = 11$ and $R = 15$ the situation is different, and it is worth being exact about the boundary. Yamamoto’s Table 3 [32, p. 45] gives, for $s = 11$, the classes $q \equiv -1, -3, -4 \equiv 7, 8, 10 \pmod{11}$: precisely the three non-residue classes in which a single prime factor already forces solvability. Theorem 1.5 supplies the complement — that $q \equiv 2$ and $q \equiv 6 \pmod{11}$ suffice as well, outside the explicitly described budget family of case (b) — and the exactness. Over the 1,005 hard primes below 4×10^5 , Yamamoto’s classes certify 413 and Theorem 1.5 certifies 628. Yamamoto does not treat $s = 15$ (his tables list prime s only), but his constructions (i)–(ii) apply verbatim there and yield the three classes $q \equiv 7, 13, 14 \pmod{15}$; Theorem 1.6 completes the Jacobi dichotomy with the fourth class $q \equiv 11 \pmod{15}$ and proves it exact (315 versus 355 primes certified in the same range). For $R \geq 19$ the divergence is no longer marginal: Yamamoto’s rules give 1, 3 or 9 classes according to $R \pmod{24}$, against the $(R-1)/2$ non-residue classes reached by the criteria and support bounds below — 3 versus 53 at $R = 107$.

Remark 1.8 (computer-assisted components). The exhaustive verifications behind Theorems 1.4, 1.5, 1.6 and Lemma 5.2 are *parts of the respective proofs*, on the same footing as a hand case analysis: each is a deterministic enumeration of a finite configuration space whose finiteness and sufficiency are proved in advance (Theorem 1.3; for Lemma 5.2, Lemma 5.1). The checks use exact integer arithmetic only and run in seconds, up to 331 s for $R = 107$; Table 5 records the configuration counts and entry points. The enumerations at $R = 7, 11, 15$ and the $R = 19, 23, 31$ cases of Lemma 5.2 are additionally formalized in Lean 4, with the reductions they rely on proved as theorems rather than trusted; Appendix A gives the inventory and the exact trust base.

The obstruction framework of Section 2 yields a reciprocity identity that converts the per-residual conditions into a single global statement about p — a structural interpretation of joint failure across residuals:

Theorem 1.9 (reciprocity structure). *Let p be a hard prime, $R \equiv 3 \pmod{4}$ prime, and q an odd prime factor of $a = (p + R)/4$ (necessarily $q \neq R$). Then*

$$\left(\frac{q}{R}\right) = \left(\frac{p}{q}\right).$$

The sieve consequences need one input per residual: a statement that failure at R forces $(p + R)/4$ to avoid prime factors in a positive-density set of residue classes. For $R = 3, 7, 11, 15$ the exact criteria provide this; for every other residual it is the support bound.

Theorem 1.10 (support bound). (i) *Let $d \geq 4$ be even and $S \subseteq \mathbb{Z}/d \setminus \{0\}$. If the bounded subset-sum set $M(S) = \{\sum_{v \in S} e_v v : e_v \in \{0, 1, 2\}\}$ is not all of $\mathbb{Z}/d$, then $|S| \leq d/2 - 1$, and the bound is attained ($S =$ the nonzero even classes).* (ii) *Consequently, for every prime $R \equiv 3 \pmod{4}$ and every p : if the prime factors of $(p + R)/4$ occupy at least $(R-1)/2$ nonzero unit classes mod R , then every unit class is a divisor class of m^2 and (1.2) holds; failure forces support $\leq (R-3)/2$ off the identity class, with no upper limit on R .* (iii) *In a finite abelian group G of even order g with t involutions (multiplicative form, $M(S) = \prod_{v \in S} \{1, v, v^2\}$): $M(S) \neq G$ implies $|S| \leq \max(\lfloor (g + t - 2)/2 \rfloor, g/2 - 1)$. For $G = (\mathbb{Z}/R)^*$, $t = 2^{\omega(R)} - 1$: failure at an admissible composite R forbids at least $\varphi(R)/2 - 1$ unit classes when $\omega(R) \leq 2$ — sieve*

dimension $\geq \frac{1}{2} - 1/\varphi(R)$ — and at least $\varphi(R)/2 - 2^{\omega(R)-1} - 1$ in general, which since $2^{\omega(R)} = R^{o(1)}$ is dimension ≥ 0.44 for every admissible R beyond an absolute bound: every residual, with no restriction on ω , is a usable sieve input. For the admissible composites $R \leq 107$ an exhaustive enumeration (`kneser_support_general`, a proof component of this clause) sharpens the maximal failing support to $g/2 - 1$, so those residuals join the chain with full dimension $\geq \frac{1}{2}$ each.

Lemma 5.2 below is the per-residual machine-verified special case in which the bound was originally established, and remains the form carried by the Lean development. Stacking these inputs across a list of residuals multiplies the savings:

Theorem 1.11 (the chain). *The number of hard primes $p \leq x$ failing every residual in $\{3, 7\}$, in $\{3, 7, 11\}$, in $\{3, 7, 11, 15\}$, and in $\{3, 7, 11, 15, 19\}$ is $O(x/(\log x)^2)$, $O(x/(\log x)^{5/2})$, $O(x/(\log x)^3)$ and $O(x/(\log x)^{7/2})$ respectively. More generally, for every finite set $\mathcal{P}$ of admissible residuals the count is $O_{\mathcal{P}}(x/(\log x)^{1+\kappa(\mathcal{P})})$ with $\kappa(\mathcal{P}) = \sum_{R \in \mathcal{P}} \delta_R$, where $\delta_R = \frac{1}{2}$ for R prime or composite ≤ 107 and $\delta_R = \frac{1}{2} - 1/\varphi(R)$ for composite $R > 107$ with $\omega(R) \leq 2$. For the fifteen primes up to 107 the exponent is $17/2$; for the full 27-element admissible list, $29/2$. Adjoining the two identity families of Proposition 1.12 raises these to $19/2$ and $31/2$.*

Proposition 1.12 (aggregate identity families). *Let $p \equiv 1 \pmod{4}$ be prime and $R \equiv 3 \pmod{4}$. If $R \mid p+1$ then $k = ap^2$ is a certificate for residual R ; if $R \mid p+4$ then $k = a^2p$ is; and no divisor shape $k = a^j p^i$ yields a further p -only condition. Hence any prime factor $\equiv 3 \pmod{4}$ of $p+1$ or of $p+4$ yields an explicit solution.*

Theorem 1.13 (density reduction). *For every $A > 0$ there is a finite admissible set $S(A)$ such that the number of hard primes $p \leq x$ for which every residual in $S(A)$ fails is $O_A(x/(\log x)^{1+A})$.*

Remark 1.14 (relation to Vaughan’s bound). As exceptional-set estimates, Theorem 1.13, the chain of Theorem 1.11, and Theorem 5.5 are all weaker than Vaughan’s bound $\ll x \exp(-c(\log x)^{2/3})$ [30], which already exceeds every fixed power of $\log x$. They are not offered as improvements on it, and the honest comparison needs one further step, because Vaughan’s argument is *constructive*: it proceeds by covering congruences, and every n it certifies comes with an explicit solution and hence with an explicit admissible residual, bounded by a power of his sifting parameter. Read that way it already gives an almost-all bound on $R_{\min}$ — roughly $R_{\min}(n) \leq \exp(O((\log x)^{1/3}))$ outside a set of size $x \exp(-c(\log x)^{2/3})$, and $R_{\min}(n) \leq (\log x)^\theta$ outside $x \exp(-c(\log \log x)^2)$ when the threshold is set to match Theorem 5.5’s exceptional set. So Theorem 5.5 is not the first almost-all bound on $R_{\min}$. Its content is that at comparable exceptional-set quality it lowers the threshold from a power of $\log x$ to a constant multiple of $\log \log x$ — an exponential gain in the certified residual, which is the quantity Proposition 1.1 shows *is* the conjecture at p . Two further differences of kind: the exceptional sets here are cut out by finitely many *explicit* multiplicative conditions on shifted forms, each failure constructively certified by an exhibited divisor k , with the underlying lemmas exact or machine-verified; and the certificates are of a single controlled shape, indexed by R , rather than assembled from a covering system. A mean-value argument, by contrast, counts the primes that fail and produces no residual at all for the ones that succeed.

Organization. Section 2 develops the obstruction framework (Propositions 2.1, 2.2) and proves Theorem 1.9, whose Corollary 2.3 depends on the character/II dichotomy introduced there. Section 3 proves Theorem 1.2, then Theorem 1.3, then Theorems 1.4, 1.5 and 1.6, which

rest on it. Section 4 is the census: data, verification protocol, the record, and the calibrated model. Section 5 is the structural context — Section 5.1 proves Theorem 1.11 (base case first, then, after the monotonicity reduction (Lemma 5.1), Lemma 5.2 and its unconditional upgrade Theorem 1.10, the higher exponents), followed by Proposition 1.12 and Theorem 1.13; Section 5.2 proves the uniform chain (Theorem 5.5); Section 5.3 localizes what stands between it and a polylogarithmic bound; and Section 5.4 sketches, in compressed form, the one conditional program that would pass the ceiling.

Table 1 records, for each main result, exactly what its proof consumes and its formal-verification status, so that each layer can be audited independently.

Three results are retained although later ones subsume them: Theorem 1.13 (superseded by Theorem 5.5, but its fixed-list proof is the template the uniform chain scales), Lemma 5.2 (subsumed by Theorem 1.10, retained as the machine-verified confirmation and the form the Lean development carries).

The census of Section 4 also calibrates the base-case constant and the decay exponents of the chain, and it is where the per-residual criteria of Section 3 are exercised at scale.

2 The obstruction framework

Proposition 2.1 (character obstruction). *Let $r \mid R$ be prime with $r \equiv 3 \pmod{4}$. If every prime factor q of $m = pa$ satisfies $\left(\frac{q}{r}\right) = +1$, then (1.2) fails for R .*

Proof. Every divisor k of m^2 is a product of prime factors of m , so $\left(\frac{k}{r}\right) = +1$. A certificate requires $k \equiv -m \pmod{r}$, whence $\left(\frac{k}{r}\right) = \left(\frac{-m}{r}\right) = -1$ by (1.4) — impossible. $\square$

We call failures produced by Proposition 2.1 *character failures*; failures where m possesses a non-residue prime factor modulo every relevant r , yet no divisor of m^2 attains the exact class $-m \pmod{R}$, are *budget failures*. (We avoid “character failure/II”, which Elsholtz and Tao [4] use for the two principal *solution* parametrizations.)

Proposition 2.2 (guaranteed-success classes). *Let $t \equiv -4^{-1}p^2 \pmod{R}$. If a has a prime factor q with $q \equiv tp^{-i} \pmod{R}$ for some $i \in \{0, 1, 2\}$, then $k = qp^i$ is a certificate. In particular, since $tp^{-2} \equiv -4^{-1} \pmod{R}$, any prime factor of a in the fixed class $-4^{-1} \pmod{R}$ certifies success regardless of p .*

Proof. $k = qp^i$ divides m^2 (as $q \mid a$ and $i \leq 2$) and lies in class $t \equiv -m$. Integrality of b, c is automatic from $k \equiv -m$ and $m^2/k \equiv m^2(-m)^{-1} \equiv -m \pmod{R}$. $\square$

The contrapositive of Proposition 2.2 is the sieve input for Theorem 1.13: failure at R forces $(p + R)/4$ to avoid prime factors in up to three explicit classes modulo R .

The next identity converts the per-residual character conditions into a single global statement about p , collapsing joint failure across residuals into one set of quadratic-character conditions.

Proof of Theorem 1.9. From $q \mid p + R$ we get $R \equiv -p \pmod{q}$. By quadratic reciprocity with $R \equiv 3 \pmod{4}$, $\left(\frac{q}{R}\right) = (-1)^{\frac{q-1}{2}} \left(\frac{R}{q}\right) = (-1)^{\frac{q-1}{2}} \left(\frac{-1}{q}\right) \left(\frac{p}{q}\right) = \left(\frac{p}{q}\right)$. $\square$

Corollary 2.3 (forced primes and correlated failure). *(i) The hard classes are the squares mod 840, so p is automatically a quadratic residue mod 3, 5, 7, and $p \equiv 1 \pmod{8}$ handles $q = 2$ (which divides a only for $R \equiv 7 \pmod{8}$, where $\left(\frac{2}{R}\right) = +1$): no prime forced into a shifted value*

result	proof inputs	proved in	machine-checked?
Prop. 1.1	elementary	§1.1	Lean, both directions
Props. 2.1, 2.2	elementary (characters; explicit k)	§2	Lean
Thm. 1.2 ($R=3$)	elementary	§3	Lean, both directions
Thm. 1.3	elementary	§3	model + reduction in Lean
Thm. 1.4 ($R=7$)	Prop. 2.1 + Thm. 1.3 + enumeration (1,536)	§3	Lean (<code>native_decide</code>)
Thm. 1.5 ($R=11$)	Thm. 1.3 + enumeration (497,664)	§3	Lean (<code>native_decide</code>)
Thm. 1.6 ($R=15$)	Thm. 1.3, composite form + enumeration (349,920)	§3	Lean (<code>native_decide</code>)
Thm. 1.9	quadratic reciprocity	§2	Lean
Lem. 5.2	Thm. 1.10; independently, Lem. 5.1 + enumerations / DP	§5.1	Lean: as a corollary of Thm. 1.10 at every prime R ; directly at $R = 19, 23, 31$
Thm. 1.10	Kneser’s theorem + Lem. 5.1; (iii) upgrade at $R \leq 107$: enumeration	§5.1	Lean, parts (i) and (iii); (iii) upgrade: Python
Thm. 1.11	Thms. 1.2, 1.4–1.6, 1.10 + Fundamental Lemma	§5.1	no
Prop. 1.12	elementary + Thm. 1.11	§5.1	families in Lean
Thm. 1.13	Prop. 2.2 + Fundamental Lemma	§5.1	no
Thm. 5.5	Thm. 1.10(iii), general tier + Brun’s sieve, uniform constants	§5.1	no
branch structure of §5.3	Kneser + Lem. 5.1 + quotient induction; complete-partition construction	§5.3	Lean (kernel step; branch count; puncture lemma)
Thm. 5.8	Conjecture 5.7 + the audit of Thm. 5.5	§5.3	no (conditional)
Lem. 5.10	Jacobi reciprocity, elementary	§5.4	Lean
Thm. 5.9	norm-form bridge (5.1) + [19], [17] + Fundamental Lemma	§5.4	no
Lem. 5.10	Jacobi reciprocity, elementary	§5.4	Lean

Table 1: Audit map: what each main result depends on, where it is proved, and its formal-verification status. “—” means the result is elementary enough that no machine check was warranted; “no” means the proof is by hand only. Theorem 5.9 is established in outline (proof sketch).

$(p + R)/4$ by the congruences (H1)–(H3) carries a negative character. (ii) p fails character at every prime residual $R \leq B$ simultaneously if and only if $(\frac{p}{R}) = +1$ for each such R and $(\frac{p}{q}) = +1$ for every unforced odd prime q (one not forced by the congruences of (H1)–(H3)) dividing any of the shifted values $(p + R)/4$, $R \leq B$ — one condition per distinct prime, shared across every residual it divides.

Proof. Immediate from Theorem 1.9 together with (H1)–(H3) and the quadratic characters of the forced small primes. $\square$

In words, part (i) says that the small primes forced into the shifted values by the Mordell structure never break a character-failure — a conceptual reading of why the hard classes are exactly the square classes. By part (ii), failures at distinct residuals correlate through shared prime factors: shifted values with few distinct prime factors leave few independent sign conditions. Section 4 quantifies this for the record prime. Both parts are verified at scale (135,588 instances of Theorem 1.9, 61,438 forced-prime instances, zero violations of either; Appendix A).

3 Exact criteria

The individual criteria proved in this section are instances; the conceptual point is the framework behind them — every residual, prime or composite, admits a finite computable exact criterion (Theorem 1.3), and every criterion doubles as a sieve input. The four residuals worked out here are the ones the computations show carry 95% of the hard primes.

3.1 Proof of Theorem 1.2

Proof. Modulo 3 the unit group has order 2, so quadratic character equals class. Since $p^2 \equiv 1 \pmod{3}$, (1.3) gives $m \equiv 1$ and the target is $-m \equiv 2 \pmod{3}$; also $3 \nmid a$. *Necessity:* if every prime factor of a is $\equiv 1 \pmod{3}$, then $a \equiv 1 \pmod{3}$, hence $p = 4a - 3 \equiv 1 \pmod{3}$ as well, so every prime factor of $m = pa$ is $\equiv 1 \pmod{3}$ and every divisor of m^2 is $\equiv 1 \pmod{3}$: class 2 is never attained. *Sufficiency:* $k = q \equiv 2 \pmod{3}$ divides m^2 , and the recovery formulae give integers as in Proposition 2.2. $\square$

3.2 Proof of Theorem 1.3

Proof. A certificate is a divisor $k \mid m^2$ in class $-m \pmod{R}$. The set of classes of divisors of m^2 is determined by the classes of the prime factors of $m = pa$ together with their multiplicities, each factor being usable with exponent at most twice its multiplicity; exponents are only meaningful modulo $\text{ord}_R(\text{class}) \leq R - 1$, so multiplicities beyond $\lceil (R - 1)/2 \rceil$ are equivalent to that cap (Remark 3.1 illustrates why). The target class and the consistency relation $\prod(\text{classes}) \equiv a \equiv 4^{-1}p \pmod{R}$ are functions of $p \pmod{R}$ by (1.3). Hence solvability is a function of the capped class multiset and $p \pmod{R}$, a finite amount of data. $\square$

Remark 3.1 (why the cap works, and the size of the space). The one step above that is not pure bookkeeping is the multiplicity cap, and a small example shows both why it is harmless and what it buys. Take $R = 7$ and a factor of a in a class x of order 6 (say $x = 3$): a factor of multiplicity μ may enter a divisor of m^2 with any exponent $0, 1, \dots, 2\mu$, and once $2\mu \geq 6$ these powers already sweep out the full cyclic group $\langle x \rangle$ — raising μ past $\lceil (R - 1)/2 \rceil = 3$ enlarges nothing that a divisor subproduct can reach. For classes of smaller order the cap is generous

rather than tight (a class of order 2 stabilizes at $\mu = 1$), which is fine: the enumeration ranges over a finite over-approximation. What finiteness buys is concrete: at $R = 7$ the capped space has 4^6 multisets before the structural constraints (i)–(iii) in the proof of Theorem 1.4 below and 1,536 consistent configurations after them; at $R = 11$ it compresses to 25 equivalence states. The enumeration is decidable, not merely heuristic, precisely because the cap and the consistency relation are proved before any code runs.

3.3 Proof of Theorem 1.4

Proof. $(\Rightarrow)$ is Proposition 2.1 with $r = R = 7$. $(\Leftarrow)$ reduces to a finite verification via Theorem 1.3. Write $(\mathbb{Z}/7)^* \cong \mathbb{Z}/6$ in discrete-log coordinates. Three structural constraints hold at hard primes:

- (i) $p \bmod 7 \in \{1, 2, 4\}$, by (H3);
- (ii) $2 \mid a$: by (H1), $p + 7 \equiv 0 \pmod{8}$, so $a = (p + 7)/4$ is even — the class of 2 (a residue mod 7) is always present;
- (iii) the factor classes multiply to $a \equiv 4^{-1}p \pmod{7}$.

Enumerating all class multisets with multiplicities capped at 3 subject to (i)–(iii) yields 1,536 consistent configurations; for each, one checks whether the target class is attained by a bounded-exponent subproduct. The verification (code: `theory.verify_R7_finite`) confirms: the target is attainable precisely when some factor class is a non-residue. $\square$

Remark 3.2. The hard-class hypotheses are essential: for $p = 701 \equiv 1 \pmod{4}$ (not hard), $a = 177 = 3 \cdot 59$ has the non-residue factor 3, yet $R = 7$ fails. Constraint (ii) — the forced even factor — is exactly what rescues every would-be budget configuration at hard primes. Empirically (Section 4), among 41,421 sampled failures of $R = 7$ at hard primes, *all* are character failure.

3.4 Proof of Theorem 1.5

Proof. By Theorem 1.3 with $R = 11$ (the unit group has order 10): multiplicities cap at 5, and at hard primes $3 \mid a$ (from (H2), $11 \equiv 2 \pmod{3}$), while $2 \nmid a$ (from (H1), $p + 11 \equiv 4 \pmod{8}$) and $7 \nmid a$ (from (H3), $-11 \bmod 7 = 3 \notin \{1, 2, 4\}$). A dynamic-programming enumeration over the finite configuration space (code: `theory.finite_criterion_dp`) compresses it to 25 equivalence states: 16 succeed, 6 fail with all factor classes quadratic residues (case (a)), and exactly 3 fail with a non-residue present (case (b)); the three failing states decode to the three listed patterns. Two structural facts emerge from the enumeration. First, no state fails by a *proper-subgroup* obstruction: the only subgroup of $\mathbb{Z}/10$ containing an odd element properly is $\{0, 5\}$, and the consistency relation forces p 's class to extend it to all of $\mathbb{Z}/10$. Second, the case-(b) failures are pure *budget* failures — the exponent bounds, not the character obstruction, block the target: any of $v_3(a) \geq 2$, a factor in one of the classes 4, 5, 9, or a second non-residue factor restores solvability. $\square$

3.5 Proof of Theorem 1.6

Proof. $(\Rightarrow)$, the Jacobi obstruction.) By (1.3), $m \equiv 4^{-1}p^2 = (2^{-1}p)^2 \pmod{15}$ is the square of a unit, so the Jacobi symbol satisfies $\left(\frac{m}{15}\right) = +1$; and $\left(\frac{-1}{15}\right) = \left(\frac{-1}{3}\right)\left(\frac{-1}{5}\right) = -1$. Hence $\left(\frac{-m}{15}\right) = -1$.

The classes with $(\frac{q}{15}) = +1$ are exactly $\{1, 2, 4, 8\} = \langle 2 \rangle$, a subgroup of $(\mathbb{Z}/15)^*$ containing p 's class ($p \bmod 15 \in \{1, 4\}$, since $p \equiv 1 \pmod{3}$ by (H2) and $p \equiv \pm 1 \pmod{5}$ for all six hard classes). If every prime factor of a lies in $\langle 2 \rangle$, then every divisor of m^2 does, and the target $-m \equiv 11 \pmod{15}$ (an equality valid for both p -classes) is unreachable. This composite-modulus form of Proposition 2.1 is strictly stronger than the proposition applied to the single prime $r = 3 \mid 15$.

($\Leftarrow$.) By the composite form of Theorem 1.3 over $(\mathbb{Z}/15)^* \cong \mathbb{Z}/2 \times \mathbb{Z}/4$ (Carmichael exponent $\lambda(15) = 4$, so reachability saturates at multiplicity 2; multiplicities 0..5 capture every pair of saturation state and class product). The structural constraints at hard primes are: $p \bmod 15 \in \{1, 4\}$; $2 \mid a$ (from (H1), $8 \mid p + 15$), so the class of 2 always appears; and consistency $\prod(\text{classes}) \equiv a \equiv 4p \pmod{15}$. Enumerating the full space (code: `theory.verify_R15_finite`) yields 349,920 consistent configurations: 349,380 succeed and 540 fail — and *every* failing configuration has all factor classes in $\langle 2 \rangle$. In contrast to $R = 11$ there are no budget failures at all: the consistency relation forces $\prod(\frac{q}{15})^{\mu_q} = (\frac{4p}{15}) = +1$, so non-residue factors occur with even total multiplicity, and the enumeration confirms that any two non-residue units (or one squared), together with the forced powers of 2, always reach the target. $\square$

Remark 3.3. The general composite- R engine (`theory.solvable_exact_general`), which evaluates the criterion of the composite finite reduction directly in $(\mathbb{Z}/R)^*$, agrees with ground-truth divisor search on *every* composite admissible residual $R \in \{15, 27, 35, \dots, 99\}$ (7,938 sampled hard primes per residual below 10^9 , zero disagreements).

4 The census to 10^{12}

This section is the paper's main object: the value of $R_{\min}(p)$ for every hard prime below 10^{12} . For each p the admissible residuals are tried in increasing order; at each R the criterion is the exact one of Section 3 where available, and otherwise a direct integer factorization of $a = (p + R)/4$ (trial division; the factorization of $m^2 = p^2 a^2$ follows) with a scan of the divisor classes for (1.2). The first R that succeeds is $R_{\min}(p)$, and the certificate it yields is re-verified in exact integer arithmetic as $4abc = p(bc + ac + ab)$. All code and data are in the accompanying repository; the 10^{12} run is a single 10.5 h job on 32 cores, reproduced by `scripts/run_1e12.sh`.

Remark 4.1 (data revision in progress). A defect was found in the factorization routine used for the 10^{12} run *after* the results below were first recorded, and this section is being revised against a corrected recomputation. The routine factors $a = (p + R)/4$ by trial division against a fixed table of primes below 180,000; that certifies the surviving cofactor prime only while $a < 180,000^2$, i.e. while $p < 1.296 \times 10^{11}$. Above that cliff a cofactor which is a product of two primes both beyond the table was recorded as prime, so the divisor set of m^2 came out incomplete.

The error is *one-directional*, and that bounds the damage exactly. Every divisor the routine does find is genuine and every certificate is exact-checked, so a residual it reports as working really does work; what it can do is fail to notice that a *smaller* residual also works. Hence the recorded $R_{\min}$ is $\geq$ the true $R_{\min}$ at every prime, with equality guaranteed below the cliff. Three consequences. (i) Existence is untouched: every hard prime below 10^{12} has an exhibited, exactly verified certificate, and `num_unsolved = 0` stands. (ii) The record stands, and in fact is now *confirmed*: the maximum of the true $R_{\min}$ is at most the maximum of the recorded values, which is 111; and it is attained, because the prime realizing 111 lies at 1.199×10^{11} , below the cliff, where the factorization is complete — so $\max R_{\min} = 111$ exactly over hard primes below 10^{12} .

What is not yet confirmed is the *number* of primes attaining it; the two others recorded at 111 lie above the cliff. (iii) The distribution of Table 3 and the deep-tail counts are affected above the cliff, which is 86% of the range: a window census at 10^{12} finds 31 of 2,282 hard primes (1.4%) with an overstated residual, so of order 10^7 entries are expected to move — always downward, from the tail toward the head. Everything at or below 10^{11} is unaffected, the cliff lying above it; that includes the 10^9 solvability masks and hence the calibration of Section 4.3, the failure taxonomy, and the band-filling test of the model, whose primes all lie in $(1.3 \times 10^{10}, 10^{11})$. The defect is fixed in the accompanying repository (`bulk_generate.factorize` now completes the cofactor by Miller–Rabin and Pollard-rho, with a regression test on the primes that exposed it), and the 10^{12} census is being regenerated. Figures marked *provisional* below are the ones awaiting that recomputation.

4.1 Verification protocol

The claims of this section rest on the datasets of Table 2, all archived in the repository under `data/` with sha256 manifests. A certificate (R, a, b, c) for p is *accepted* only if $4abc = p(bc+ac+ab)$ holds in exact integer arithmetic and $R = 4a - p$; for the compact minimal- R maps, the verifier *reconstructs* the triple from (p, R) by the divisor search of (1.2) and then applies the same exact check, so a map entry is verified as a genuine certificate, not taken on faith.

The masks dataset of Table 2 is the basis for the covering-set statements (the 18-residual cover, the lower bound 12, the four critical primes), the failure taxonomy, and the calibration below; the minimal- R maps are the basis for Table 3, the record, and the gap analysis.

4.2 Data

Every one of the 1,175,215,396 hard primes $p < 10^{12}$ possesses a residual certificate with $R \leq 111$. The verification behind this claim is graded by dataset: exhaustive reconstruction and exact re-checking of every entry below 10^9 ; systematic samples (143,000, 257,343 and 2,350,431 entries) at 10^{10} , 10^{11} and 10^{12} , together with the *complete* tails (842 entries with $R \geq 51$, 20,151 and 118,210 with $R \geq 43$) re-checked with full minimality — Table 2 gives the per-dataset detail. The minimal residuals distribute as in Table 3 and Figure 1.

Three features of the distribution deserve comment.

- **The maximum moves at 10^{12} .** Throughout $10^7 \leq x \leq 10^{11}$ — five successive orders of magnitude — the maximal minimal residual over hard primes $p \leq x$ stood at $R = 107$, attained by the single prime $8,803,369 < 10^7$. At 10^{12} it is $R_{\max} = 111$ — confirmed exactly, by Remark 4.1(ii) — recorded as attained by three primes (a count still to be confirmed), the least being

$$p = 119,945,383,009 \equiv 529 \pmod{840},$$

for which every admissible $R \leq 107$ fails and $R = 111$ certifies with $a = (p + 111)/4 = 29,986,345,780$. The old record 107 also ceased to be unique, acquiring three further primes in the same decade. The prime 8,803,369 is extremal for a closely related statistic as well: in independent contemporaneous work, Bello-Hernández, Benito and Fernández [10] report it as the largest “survivor” below 10^7 in a $k \equiv 3 \pmod{4}$ divisor sieve, requiring $k = 107$ there too. In the language of Corollary 2.3, 8,803,369’s 27 shifted values contain only 43 distinct odd primes (typical: 45–52), and 81% of them satisfy $(\frac{p}{q}) = +1$ (typical: 44–60%) — an extreme value of exactly the statistic Theorem 1.9 identifies as governing joint

dataset	entries	verification performed
explicit triples, $p < 1.2 \times 10^8$	213,131	every triple checked exactly
minimal- R map, $p < 10^9$	1,587,581	<i>exhaustive</i> : every entry reconstructed and checked exactly
minimal- R map, $p < 10^{10}$	14,215,707	143,000 systematically sampled entries reconstructed and checked; additionally <i>all</i> 842 tail entries ($R \geq 51$) checked with full minimality (no smaller admissible R succeeds)
solvability masks, $p < 10^9$	$27 \times 1,587,581$	per-prime solvability of every admissible $R \leq 107$ by full divisor-class computation; consistency with the minimal- R map
R -sequence dataset, $p < 10^{11}$	128,671,219	257,343 systematically sampled entries reconstructed and checked; <i>all</i> 20,151 tail entries ($R \geq 43$) checked with full minimality; prime array pinned by sha256; pipeline validated byte-for-byte against the exhaustively verified 10^9 map
R -sequence dataset, $p < 10^{12}$	1,175,215,396	2,350,431 systematically sampled entries reconstructed and checked; <i>all</i> 118,210 tail entries ($R \geq 43$) checked with full minimality — but see Remark 4.1: above 1.296×10^{11} the minimality re-check shared the defective factorization and so is not independent of it; it remains a valid check of the certificate itself. Included are the three entries at $R = 111$; zero unsolved; both arrays pinned by sha256; single 10.5 h run on 32 cores, regenerated by <code>scripts/run_1e12.sh</code>

Table 2: Certificate datasets and the verification applied to each.

failure. We attach no significance level to this: the prime was selected *because* it was the record, so its extremeness in a correlated statistic is expected rather than surprising. The observation is descriptive — it says what a record prime looks like mechanically, not that its existence is improbable.

- **A prediction, and what it was worth.** The *calibrated independence model* treats the per-residual failure events as independent across R , with per-residual failure rates fitted from the complete 10^9 solvability masks and a single deep-tail correlation factor calibrated on the $R > 83$ counts; it is a heuristic, not a theorem, and its role is to make falsifiable predictions. Below 10^{10} no prime had minimal $R \in \{87, 91, 95, 99, 103\}$; the model attributed this gap to small-number statistics (expected counts ≤ 0.12 per value at 10^9) and predicted that deep-tail primes (failing every $R \leq 83$) land at 87, 91, 95, 99, 103, ≥ 107 with probabilities 0.53, 0.11, 0.22, 0.03, 0.05, 0.06 (the last being the total mass at or beyond 107). Between 10^{10} and 10^{11} the five values acquired 8, 3, 5, 1, 1 primes respectively

R	count	R	count	R	count
3	657,645,152	43	43,558	83	78
7	286,441,544	47	50,190	87	40
11	167,419,973	51	8,756	91	5
15	33,222,834	55	7,441	95	12
19	19,307,426	59	4,863	99	4
23	7,996,742	63	1,439	103	5
27	1,376,271	67	488	107	4
31	1,318,258	71	970	111	3
35	208,713	75	146		
39	160,273	79	208		

Table 3: Distribution of the minimal residual $R_{\min}$ over the 1,175,215,396 hard primes below 10^{12} (plotted in Figure 1). The value $R = 111$ is new at this scale. *Provisional above $p = 1.296 \times 10^{11}$* : see Remark 4.1. Counts there are upper bounds for the tail and lower bounds for the head; the totals and the maximum are unaffected.

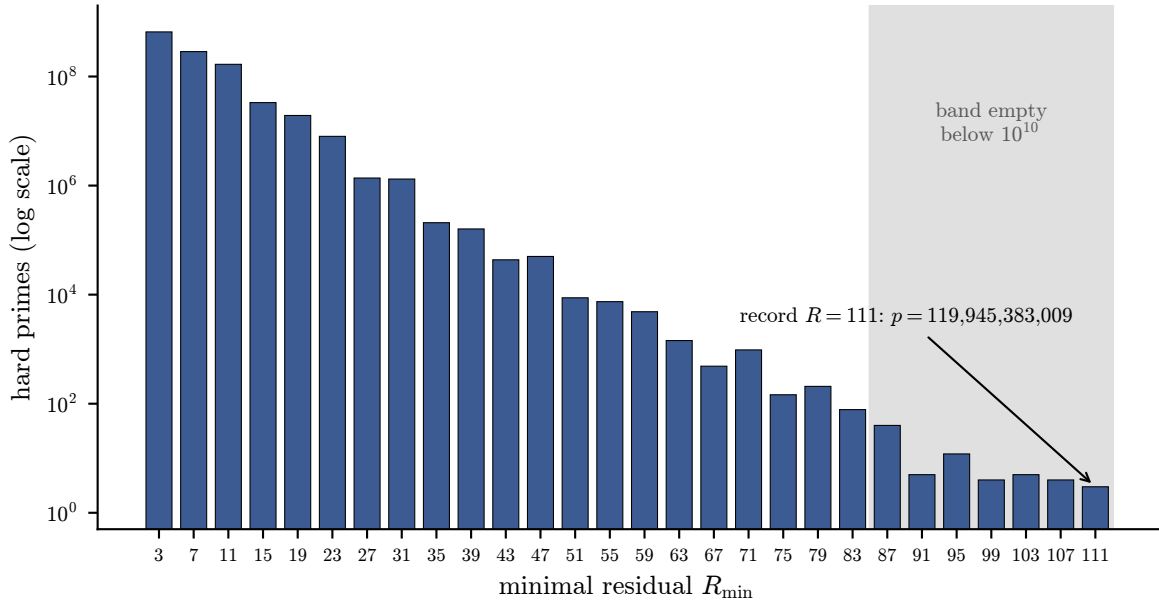


Figure 1: The data of Table 3 on a logarithmic count scale. The head is dominated by $R = 3, 7, 11$; the tail thins geometrically; the shaded band 87–111 was empty below 10^{10} , and its population at 10^{11} and 10^{12} followed the calibrated prediction (see the prediction bullet below), including the move of the maximum from $R = 107$ to $R = 111$.

(all in $(1.3 \times 10^{10}, 10^{11})$), and none passed 103; that distribution was *consistent* with the prediction ($\chi^2 \approx 1.3$) but, on 19 points with several expected cells below 5, had little power to discriminate. The model's sharper commitment was about *when the record would move*: the mass at or beyond 107 was put at 6% per deep-tail prime, which with the deep-tail growth rate places the first record-breaking prime in the decade 10^{11} – 10^{12} . It broke there,

at $R = 111$, at a prime below the cliff of Remark 4.1 and hence correctly computed. The deep tail is recorded as holding 73 primes with $R_{\min} \geq 87$ against 19 a decade earlier, distributed 40, 5, 12, 4, 5, 4, 3 over 87, $\dots$, 111 — *provisional*: 22 of the 73 lie below the cliff and are confirmed, the rest await recomputation and can only move downward. A model fitted at 10^9 thus predicted, before the computation, both the filling of a band that was empty at the time and the decade in which the maximum would move for the first time in five orders of magnitude.

- Among all 1,587,581 hard primes below 10^9 , exactly *four* admit a unique working residual ≤ 107 (8,803,369 with 107; 142,361,209 with 59; 287,567,281 with 83; 794,037,841 with 63) — the entire obstruction to a shorter covering list. A greedy cover of the full solvability table uses 18 residuals; a packing argument gives a lower bound of 12.

4.3 Calibration of the base case of Theorem 1.11

The proportion of hard primes failing both $R = 3$ and $R = 7$, multiplied by $\log x$, is constant to within 1% across three decades of the 10^9 data (Table 4).

bin ($p \sim$)	10^6	10^7	6×10^7	2.5×10^8	10^9
density $\times \log x$	5.14	5.19	5.18	5.17	5.16

Table 4: The proportion of hard primes near x that fail both $R = 3$ and $R = 7$, multiplied by $\log x$, in five bins spanning three decades of the 10^9 data (x is the bin’s representative value). Constancy of the product is the decay rate predicted by the base case of Theorem 1.11.

That is, the joint exceptional density is $\approx 5.17/\log x$: the decay matches the order predicted by the base case of Theorem 1.11, and 5.17 estimates the (conjectural) Selberg–Delange constant. The exceptional set is density zero but decays slowly: 26.5% of hard primes below 10^9 fail both residuals; all are settled by deeper residuals of the list. The fitted per-residual decay exponents are consistent with the sieve prediction $\kappa = \frac{1}{2}$ where exact criteria exist: $\hat{\kappa} = 0.539$ for $R = 7$, and $\hat{\kappa} = 0.387$ for $R = 3$, where the short fitting range and Selberg–Delange lower-order terms bias the exponent downward.

4.4 Failure taxonomy

Sampling the complete solvability table at 10^9 : failures at $R = 7$ are 100% character failures (as Theorem 1.4 requires); at $R = 11$, 89.7% character failure and 10.3% the budget pattern of Theorem 1.5(b); the character-failure share broadly decreases with $\varphi(R)$ (e.g. 51% at $R = 103$, though not monotonically), as hitting one specific class among $\varphi(R)$ grows harder while the character dichotomy stays binary. Taken together, the data of this section matches the framework of Sections 2–5.1: each distributional feature — the head, the geometric tail, the once-empty band, the static record — has a structural or heuristic explanation (the exact criteria for the head; the calibrated model, built on the sieve dimensions, for the tail and the band; Theorem 1.9 for the record).

5 The sieve chain and its ceiling

This section is the structural context around the census: what the per-residual criteria imply about the tail of the distribution of $R_{\min}$, how far that reasoning reaches, and where it stops. It is deliberately compressed. The results are stated with proofs where the proof is short and load-bearing, and in outline where the argument is a routine repetition of one already given; the fuller development, including the measurements quoted here, is in the accompanying repository.

5.1 The chain

Proof of Theorem 1.11, base case $\{3, 7\}$. Step 1 (exact criteria). Let p be a hard prime and set $n = (p + 3)/4$, so that $(p + 7)/4 = n + 1$: *the two shifted forms are consecutive integers.* By Theorems 1.2 and 1.4, p is exceptional (both residuals fail) precisely when

$$n \text{ has no prime factor } \equiv 2 \pmod{3} \quad \text{and} \quad n + 1 \text{ has no prime factor } \equiv 3, 5, 6 \pmod{7},$$

with $p = 4n - 3$ prime.

Step 2 (containment in a sifted set). Fix one of the six hard classes $h \pmod{840}$; then $n \equiv n_0 \pmod{210}$ is fixed and the conditions at 2, 3, 5, 7 are determined by the class (in particular $3 \nmid n$ and $7 \nmid n + 1$ automatically), so it suffices to bound each class and sum. Let $z = x^{1/10}$ and discard the $O(z)$ primes $p \leq z$. Every exceptional p then gives $n \leq (x + 3)/4$ in the class such that for every prime q with $7 < q \leq z$: $q \equiv 2 \pmod{3} \implies n \not\equiv 0$; $q \equiv 3, 5, 6 \pmod{7} \implies n \not\equiv -1$; and $q \nmid 4n - 3$, since $4n - 3 = p$ is prime and $p > z$. The first two use only “no forbidden factor up to z ”, which the full criteria imply — an upper bound needs no more.

Step 3 (sieve of dimension 2). This is a standard upper-bound sieve for the three linear forms $n, n + 1, 4n - 3$, with sifting function $\omega(q) = \mathbf{1}_{q \equiv 2(3)} + \mathbf{1}_{q \text{ NQR}(7)} + 1$ for $q > 7$, whose forbidden classes $0, -1, 3 \cdot 4^{-1} \pmod{q}$ are pairwise distinct for $q \nmid 21$. By Dirichlet, ω has average $\kappa = \frac{1}{2} + \frac{1}{2} + 1 = 2$ in the Halberstam–Richert sense $\sum_{q \leq w} \omega(q) \log q/q = \kappa \log w + O(1)$. The Fundamental Lemma of sieve theory (see [8, Thm. 2.5] and [6, Ch. 6]) with $z = x^{1/10}$ bounds the exceptional set $E(x)$ of hard primes $p \leq x$ failing both residuals:

$$\#E(x) \ll x \prod_{7 < q \leq z} \left(1 - \frac{\omega(q)}{q}\right) \asymp x (\log z)^{-1} (\log z)^{-1/2} (\log z)^{-1/2} \asymp \frac{x}{(\log x)^2},$$

by Mertens’ theorem and its arithmetic-progression form. Summing over the six classes preserves the bound. $\square$

The higher exponents need one sifting input per residual, and that is Theorem 1.10. Its proof rests on an elementary but load-bearing monotonicity, which we isolate for scrutiny.

Lemma 5.1 (monotonicity reduction). *For a class multiset C (classes v with multiplicities μ_v) let $D(C) = \{\sum_v e_v v : 0 \leq e_v \leq 2\mu_v\}$ denote the set of attainable divisor logs (before the p -shifts). If $\mu_v \leq \mu'_v$ for all v , then $D(C) \subseteq D(C')$; likewise enlarging the support enlarges D . Consequently a support S carries some failing configuration if and only if the all-multiplicity-one configuration on S fails.*

Proof. Any admissible exponent vector for C is admissible for C' , giving the inclusions. If a configuration on S with multiplicities $\mu_v \geq 1$ fails, the target lies outside $D(C)$, hence outside the subset $D(C_1)$ attained at multiplicities one — so C_1 fails too; the converse inclusion is trivial. The same argument applies verbatim after adjoining the p -shifts. $\square$

Theorem 1.10 is a statement of the critical-number literature, and we should say so before proving it. The *critical number* $\text{cr}(G)$ of a finite abelian group is the least t such that every $A \subseteq G \setminus \{0\}$ with $|A| \geq t$ has $\Sigma(A) = G$, the set of subset sums. Diderrich and Mann [13] proved $\text{cr}(G) = |G|/2$ for every abelian G of even order, with exactly four exceptions — $\mathbb{Z}/4$, $\mathbb{Z}/6$, $\mathbb{Z}/8$ and $\mathbb{Z}/2 \times \mathbb{Z}/4$ — and the determination for all finite abelian groups was completed by Freeze, Gao and Geroldinger [14]; see also [7]. Since $M(S) = \Sigma(S \uplus S) \supseteq \Sigma(S)$, parts (i) and (iii) below follow from those theorems wherever they apply, the extremal family (the nonzero even classes) is theirs, and this whole section lives inside the zero-sum circle of ideas (see [12, Ch. 6, 16]). Two things justify giving the proof anyway. First, the exceptions have to be handled, and the exponent budget removes them: among admissible R the exceptional groups occur at exactly $R = 7$ ($\mathbb{Z}/6$) and $R = 15$ ($\mathbb{Z}/2 \times \mathbb{Z}/4$), no admissible R having $\varphi(R) = 4$; there $|S| \leq g/2 - 1$ is false for $\Sigma(S)$ but true for $M(S)$, because allowing each class twice is exactly enough to close the gap, so the sieve dimension $\frac{1}{2}$ is available at *every* admissible residual with no exceptions. (At $R = 7, 15$ this is belt and braces: Theorems 1.4 and 1.6 give an exact criterion there in any case.) Second, the derivation yields the form we need — a single prescribed target, bounds uniform in R , the involution bookkeeping of part (iii) — directly, and it is what the Lean development formalizes.

Proof of Theorem 1.10. (i) Write $M = A_1 + \cdots + A_k$ with $A_i = \{0, v_i, 2v_i\}$, $k = |S|$, and let $H = \text{Stab}(M)$, a subgroup of $\mathbb{Z}/d$. Kneser’s addition theorem [20] (in the many-summand form obtained by iterating [29, Thm. 5.5]) gives $|M| \geq \sum_i |A_i + H| - (k-1)|H|$. *Case* $H = \{0\}$. Every $|A_i| = 3$, except that the (at most one) index i with $v_i = d/2$ has $|A_i| = 2$; so $|M| \geq (3k-1) - (k-1) = 2k$. Since $M \neq \mathbb{Z}/d$, $2k \leq d-1$, and d even forces $k \leq d/2 - 1$. *Case* $|H| = h > 1$. M is a union of H -cosets and $M \neq \mathbb{Z}/d$, so $|M| \leq d-h$. Split $k_0 = |S \cap H| \leq h-1$ and $k_1 = k - k_0$. For $v \in H$, $|A_v + H| = h$; for $v \notin H$, $A_v + H \supseteq H \cup (v+H)$, so $|A_v + H| \geq 2h$. Kneser’s bound gives $d-h \geq h(k_0 + 2k_1 - k + 1) = h(k_1 + 1)$, so $k_1 \leq d/h - 2$ and $k \leq h + d/h - 3$. For every divisor h with $2 \leq h \leq d/2$ the quadratic $h^2 - (\frac{d}{2} + 2)h + d \leq 0$ (roots 2 and $d/2$), i.e. $h + d/h - 3 \leq d/2 - 1$. Tightness: for S the nonzero even classes, $M(S)$ is the even subgroup and $|S| = d/2 - 1$.

(ii) In discrete-log coordinates on $(\mathbb{Z}/R)^* \cong \mathbb{Z}/d$, $d = R-1$, the divisor classes of m^2 contain $M(S)$ over the nonzero support S of $(p+R)/4$ at exponent budget 2 per class (Lemma 5.1). If $|S| \geq d/2$, part (i) forces $M(S) = \mathbb{Z}/d$: every class — in particular the target $-m$ — is a divisor class of m^2 , without invoking the p -powers, and (1.2) holds. Contrapositively, failure at R forces $|S| \leq (R-3)/2$.

(iii) The argument in (i) never used cyclicity. In a finite abelian G of even order g with t involutions: in the H -trivial case each involution $v \in S$ has $|\{1, v, v^2\}| = 2$, so $|M| \geq 2k - t + 1$ and $M \neq G$ gives $k \leq \lfloor (g+t-2)/2 \rfloor$; the $|H| = h > 1$ case is verbatim, giving $k \leq h + g/h - 3 \leq g/2 - 1$. For $G = (\mathbb{Z}/R)^*$, $t = 2^{\omega(R)} - 1$; with $\omega(R) \leq 2$ the bound is $\leq g/2$, so failure forbids at least $\varphi(R)/2 - 1$ non-identity unit classes — dimension $\geq \frac{1}{2} - 1/\varphi(R)$. Without the specialization, failure forbids at least $(g-t)/2$ non-identity classes (minus one when the max is attained by $g/2 - 1$), and $2^{\omega(R)} = R^{o(1)}$ gives the stated 0.44 threshold beyond an absolute bound. The upgrade to $\varphi(R)/2$ forbidden classes for $R \leq 107$ is the enumeration cited in the statement: the maximum support of a proper product-set is exactly $g/2 - 1$ for every composite admissible $R \leq 107$ (`theory.kneser_support_general`; e.g. $R = 95$: $g = 72$, maximum 35 over 583,455 realizable product-sets). That enumeration is a proof component of the $R \leq 107$ clause; the $\frac{1}{2} - 1/\varphi(R)$ tier needs Kneser alone. $\square$

Lemma 5.2 (support bound; verified for every prime $R \equiv 3 \pmod{4}$, $19 \leq R \leq 107$). *Let R be one of the twelve primes 19, 23, 31, 43, 47, 59, 67, 71, 79, 83, 103, 107, and call the set of unit classes mod R occupied by the prime factors of $(p + R)/4$ its support. Every configuration for which residual R fails at a hard prime has support of size at most $(R - 1)/2$, at most $(R - 3)/2$ of whose classes differ from the identity class. Consequently failure at R forbids at least half the classes, through finitely many maximal-support branches.*

Remark 5.3 (the machine-verified form). Lemma 5.2 is the special case originally established by direct verification, and remains the form the Lean development carries. By Lemma 5.1 it suffices to check, per residual and per class of $p \pmod{R}$, that every support of size $(R - 1)/2$ reaches the target at minimal multiplicities; the check ignores the consistency relation and so examines a superset of the realizable configurations. At $R = 19, 23$ direct enumeration suffices (437,580 resp. 7,759,752 pairs); beyond that it is a subset dynamic program over reachability masks (3,001 states at $R = 31$ against 77.6 million supports; 19,530,971 at $R = 107$, 331 s). All twelve verify with zero violations (Table 5).

Proof of Theorem 1.11, higher exponents; sketch. Write $n = (p + 3)/4$, so the shifted forms $n + (R - 3)/4$, $R \in \mathcal{P}$, are distinct integer shifts of n , joined by $4n - 3 = p$. Each residual contributes a sifting condition on its own form: the exact criteria at $R = 3, 7, 11, 15$, and Theorem 1.10 (part (ii) at primes, part (iii) at composites) at every other residual, in each case through finitely many maximal-support branches, each forbidding a set of prime classes of density $\geq \delta_R$. Crossing the branches leaves finitely many combined problems, each a multi-form sieve of dimension $\geq 1 + \sum_R \delta_R$, the forbidden residues being pairwise distinct for $q > \max \mathcal{P}$ and smaller primes absorbed into the congruence class of n ; the Fundamental Lemma applied to each and summed gives the stated bound, the exponent being the minimum over branches. Two branch computations are worth recording, since they show the minimum is attained by the character branch and not by a budget branch: at $\{3, 7, 11\}$, branch (a) of Theorem 1.5 has dimension $1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{5}{2}$ while the budget branch (b) forbids six classes mod 11 and has dimension $\frac{13}{5} > \frac{5}{2}$, so it is lower-order; adjoining $R = 15$ adds a single branch of density $\frac{4}{8}$, raising each combined branch to dimension ≥ 3 . $\square$

Proof of Proposition 1.12. $k = ap^2$ divides $m^2 = p^2a^2$ and $k \equiv -m$ if and only if $ap^2 \equiv -pa$, i.e. $p \equiv -1 \pmod{R}$; $k = a^2p$ divides m^2 and $a^2p \equiv -pa$ if and only if $a \equiv -1$, i.e. $p \equiv -4 \pmod{R}$ by (1.3). Integrality of b, c is automatic. Scanning all shapes $k = a^j p^i$ ($j, i \leq 2$) yields no further p -only condition. The failure of both families states that the two forms $(p + 1)/2$ and $p + 4$ (in n : $2n - 1$ and $4n + 1$) are free of primes $\equiv 3 \pmod{4}$ — two further sifting conditions of dimension $\frac{1}{2}$ each on forms distinct from all previous ones. $\square$

Empirically the two families alone cover 74% of hard primes below 10^9 — and none of the four primes of Section 4 that admit a unique working residual ≤ 107 .

Theorem 1.13 is the same argument with the list chosen to make the dimension exceed $1 + A$: take $S(A) = \{R \equiv 3 \pmod{4} : 3 \leq R \leq B\}$ with B large enough that $\sum_{R \in S(A)} 1/\varphi(R) > A$, possible since the reciprocal sum over $R \equiv 3 \pmod{4}$ diverges. Within a fixed class of n modulo $840 \prod_{R \in S(A)} R$, Proposition 2.2 makes failure at R imply that $n + (R - 3)/4$ avoids a fixed nonempty set of $t_R \in \{1, 2, 3\}$ prime classes mod R , so the sifting function has average $\kappa = 1 + \sum_R t_R/\varphi(R) > 1 + A$ and the Fundamental Lemma gives $O_A(x/(\log x)^{1+A})$.

Remark 5.4 (what upper-bound sieving cannot do). By Theorem 1.10 every residual adds $\frac{1}{2}$ to the exponent, so the chain's exponent is bounded only by the number of residuals used; the

constants $O_{\mathcal{P}}$, however, grow with $|\mathcal{P}|$ (branch counts and the remainder term of the Fundamental Lemma). Theorem 1.13 is the strongest statement of this kind: density $O((\log x)^{-A})$ for every A , but never emptiness — the count bound $x/(\log x)^{\kappa}$ diverges for every fixed κ , and extrapolating the calibration of Section 4 to any fixed list yields a positive density constant, i.e. infinitely many expected exceptional primes with the record minimal residual growing slowly without bound. The conjecture for hard primes would follow from a *finite covering hypothesis* — some finite S containing a working residual for every hard prime — and the data to 10^{12} is covered by $S = \{3, 7, 11, \dots, 111\}$; but both the calibration and the conditional construction of Section 6 predict that any fixed list eventually fails. The correctly posed remaining problem is a slowly growing bound $R_{\min}(p) \ll f(p)$, which needs existence technology beyond upper-bound sieves.

5.2 The uniform chain

Theorem 1.13 fixes the list once A is fixed; letting the list *grow with x* is purely a question of uniformity in the constants — the branch count per residual, the class modulus, and the sieve constants in growing dimension. Auditing those constants shows the growth is affordable for lists of length a small multiple of $\log \log x$, which yields the strongest unconditional statement here.

Theorem 5.5 (uniform chain). *There are absolute constants $\varepsilon, c > 0$ and an (ineffective) x_0 such that for all $x \geq x_0$,*

$$\#\{\text{hard primes } p \leq x : R_{\min}(p) > \varepsilon \log \log x\} \leq x \exp(-c(\log \log x)^2).$$

In particular, all but $O(xe^{-c(\log \log x)^2})$ hard primes $p \leq x$ possess an explicit, divisor-exhibited residual certificate at some admissible $R \leq \varepsilon \log \log x$ — unconditionally.

Proof. Write $L = \log \log x$, $B = \varepsilon L$, and let $\mathcal{R}$ be the admissible residuals $3 \leq R \leq B$ ($|\mathcal{R}| \sim B/4$). Discard the $O(\sqrt{x})$ primes $p \leq \sqrt{x}$. A hard prime with $R_{\min}(p) > B$ fails every $R \in \mathcal{R}$.

Dimensions. Each $R \in \mathcal{R}$ contributes a sifting dimension δ_R : the exact criteria at $R = 3, 7, 11, 15$ and Theorem 1.10(ii) at primes give $\frac{1}{2}$; composites ≤ 107 give $\frac{1}{2}$ by the sharpened tier of Theorem 1.10(iii); and every remaining composite — with *no* restriction on $\omega(R)$ — gives $\delta_R \geq \frac{1}{2} - (2^{\omega(R)-1} + 1)/\varphi(R) \geq 0.44$ beyond an absolute bound R_1 . Hence $\kappa := 1 + \sum_R \delta_R \geq 1 + B/10$ for B large.

Branches. For each R , failure confines the support of $(p + R)/4$ to one of at most $2^{\varphi(R)}$ failing supports S_R (crudely, all subsets); fixing $(S_R)_R$ and the class n_0 of $n = (p + 3)/4$ modulo $\widetilde{M} = \text{lcm}(840, \prod_{\ell \leq B} \ell) = e^{O(B)}$ costs $\prod_R 2^{\varphi(R)} \cdot e^{O(B)} = e^{O(B^2)}$ branches.

Sieve. Within a branch, sift the integers $n \leq (x + 3)/4$, $n \equiv n_0 \pmod{\widetilde{M}}$, by the primes $\ell \in (B, z]$, $z = x^{1/s}$: the forbidden classes mod ℓ are $n \equiv 3 \cdot 4^{-1}$ (so that $p = 4n - 3$ keeps no factor $\leq z$) and, for each R with $\ell \bmod R$ outside S_R , $n \equiv -(R - 3)/4$ — pairwise distinct for $\ell > B$, so $\omega(\ell) \leq |\mathcal{R}| + 1 < \ell/2$. The density condition holds with the dimension κ above and (Ω) -constant $O(B \log B)$, by Mertens' theorem in the progressions modulo $R \leq B$ (Siegel–Walfisz range, whence the ineffective x_0). Apply Brun's sieve of order $r \asymp \kappa$ ([8, Ch. 2]; for constants uniform in the dimension, [6, Thm. 6.9]) at $s = 10r$, so that $z^r = x^{1/10}$: since the sifted sequence consists of *integers* in a progression — no primality is detected anywhere — the remainder is $\sum_{d \leq z^r} (B + 2)^{\omega(d)} \ll x^{1/10} e^{O(BL)} = x^{1/10 + o(1)}$, and the main term is $\ll x \exp(-\kappa(L - O(\log B)) + O(B \log B))$.

Accounting. Multiplying by the branch count,

$$\#\{p \leq x : R_{\min}(p) > B\} \leq e^{O(B^2)} x e^{-\kappa L + O(B \log B) + O(\kappa \log B)} \leq x e^{-BL/10 + O(B^2)},$$

and with $B = \varepsilon L$ the exponent is $-\varepsilon L^2/10 + O(\varepsilon^2 L^2) \leq -\varepsilon L^2/20$ for ε below an absolute threshold. $\square$

Remark 5.6 (the constant, the range, and the ceiling). The constant c is not innocuous: it inherits the branch factor $e^{\Theta(B^2)}$ against the sieve gain $e^{-BL/10}$, so $c \asymp \varepsilon$ is small, and the bound beats the trivial count $\asymp x/(32 \log x)$ of hard primes only once $c(\log \log x)^2 - \log \log x > \log 32$ — at $c = 10^{-2}$, only for $\log \log x \gtrsim 350$. The theorem is asymptotic in the strict sense: nontrivial for all large x , ineffectively so (Siegel–Walfisz), and of no numerical content in any accessible range, where the picture is supplied instead by Section 4. The same branch factor is the ceiling: it forces $B \ll L$ and hence $\exp(-cL^2)$, and an audit shows it is the *only* binding wall — the class modulus $\widetilde{M} = e^{O(B)}$ never multiplies the main term (the classes partition the integers), and the Mertens/Siegel–Walfisz inputs remain valid for B up to any fixed power of $\log x$ at the already-paid price of ineffectivity. Classifying the failing supports into polynomially many structured branches is therefore the one lever, and it is the subject of Section 5.3. Vaughan’s bound [30] remains stronger as a pure density statement at every tier of this family (Remark 1.14).

5.3 What stands in the way

Fix a prime residual R , $d = R - 1$, and work in the discrete-log model of Theorem 1.10(i): a *support* $S \subseteq \mathbb{Z}/d \setminus \{0\}$ has reach $M(S) = \sum_{v \in S} \{0, v, 2v\}$, and S *fails* for a target t if $t \notin M(S)$. By Lemma 5.1 every failing configuration lives on a failing support, and every failing support extends to a maximal one. Two facts calibrate the problem.

First, enumeration is hopeless. Taking $S = [1, r] \cup T' \cup \{X\}$ with $T' \subseteq [r+1, 2r]$, $|T'| = \lceil r/2 \rceil$, $r = \lceil \sqrt{d/4} \rceil$ and X completing the sum to $d/2 - 1$ gives, for large even d , at least $2^{c\sqrt{d}}$ pairwise distinct maximal failing supports for the target $t = -1$: each is a complete partition with reach exactly $\mathbb{Z}/d \setminus \{-1\}$. The exponential-in- $\sqrt{d}$ order is the classical complete-partition phenomenon — such counts are $\exp(\Theta(\sqrt{n}))$ (Andrews–Beck–Hopkins [11]) — and what matters here is only the consequence: a useful classification must produce *containers*, polynomially many structured sets covering all failing supports, not a list.

Second, the periodic case is already under control, by the standard stabilizer/quotient reduction that drives the critical-number literature (see [12, Ch. 6]); we claim no novelty in the mechanism. If S fails for t and $H = \text{Stab}(M(S)) \neq \{0\}$ with quotient $\varphi_H: \mathbb{Z}/d \rightarrow \mathbb{Z}/\bar{d}$, then $\bar{T} = \varphi_H(S \setminus H)$ is a failing support for $\bar{t} = \varphi_H(t) \neq 0$ at the same budget, $S \subseteq (H \setminus \{0\}) \cup \varphi_H^{-1}(\bar{T})$, and $H \setminus \{0\} \subseteq S$ when S is maximal. Iterating, *every* failing support of $\mathbb{Z}/d$ sits inside $\psi^{-1}(\{0\} \cup T') \setminus \{0\}$ for a composed quotient ψ and an *aperiodic* maximal failing support T' downstairs, a container of at most $\lfloor d/2 \rfloor - 1$ elements.

So everything reduces to the aperiodic case, and all the evidence points to one container shape — a coset-progression *window*: for $K \leq \mathbb{Z}/d$, a unit direction and a circular window W of quotient classes, set $C(K, W) = \varphi_K^{-1}(\{0\} \cup W) \setminus \{0\}$. The family of such containers of size $\leq \lfloor d/2 \rfloor - 1$ has fewer than d^3 members.

Conjecture 5.7 (aperiodic branch classification). *Every maximal failing support S with $\text{Stab}(M(S)) = \{0\}$ for a target t in $\mathbb{Z}/d$ satisfies $S \subseteq C(K, W)$ for some window container with $|C(K, W)| \leq \lfloor d/2 \rfloor - 1$.*

The evidence is extensive: exhaustive verification for all moduli $d \leq 60$ and all targets (24.25 million orbit-weighted maximal supports, zero violations, the bound attained hence optimal); exhaustive verification at $d = 72$, the smallest non-Hajós cyclic order — where aperiodic factorization pathologies could first appear — over all 11 target orbits and 14.99 million maximal supports, with stratified sampling at every non-Hajós order up to 240; independent verification in the residual model at $R = 19, 23, 31, 43, 47, 59$ and at the composite moduli $R = 15, 35, 39$ (entry point `erdos_straus.branch_enum`, archived); a proved subfamily; and a rank-2 witness ($S = \{11, 17, 19, 29\} \subseteq \mathbb{Z}/30$) showing the subgroup component of the shape is necessary. Where the remaining difficulty lives is also localized: near-critical Kneser analysis is vacuous here (aperiodic failing supports with $|S| \geq 3$ are never Kneser-tight), and punctured critical deletion steps are impossible — if the two new elements of a critical step at v are isolated punctures of the complement then $F - v = F$ for the miss set F , forcing $v \in \text{Stab}(M(S))$, so in the aperiodic case every critical step is of window type. What is open is the control of *non-critical* steps.

Theorem 5.8 (conditional polylog tier). *Assume Conjecture 5.7 for all moduli. Then every failing support for (d, t) lies in one of fewer than d^3 explicit window containers of size $\leq \lfloor d/2 \rfloor - 1$, so failure at R confines the factor classes of $(p+R)/4$ to $O(R^3)$ branches rather than $2^{\varphi(R)}$; and rerunning the proof of Theorem 5.5 with list length $B = (\log x)^\theta$, there is an absolute $c_0 > 0$ such that all but $O(x \exp(-(\log x)^{c_0}))$ hard primes $p \leq x$ satisfy $R_{\min}(p) \leq (\log x)^{c_0}$.*

The balance is a sieve gain $\delta(1 - \theta)L$ against a branch-and-slop cost $(3 + c_M)\theta L$ with $\delta = \frac{1}{2}$ and $c_M = O(1)$ collecting the Mertens-in-progressions constants, the discarded initial segment of sieving primes, and the (Ω) -condition constant; this is admissible for $\theta < \delta/(3 + c_M + \delta)$, roughly $1/9$ with the constants proved here. The family ceiling sits strictly below $\frac{1}{2}$: no container scheme of this shape has fewer than $\sim d$ members, so the count exponent is at least 1 and θ is capped below $\frac{1}{3}$ — short of Vaughan’s $2/3$. Conjecture 5.7 is thus a single well-posed question of additive combinatorics, a container theorem of Kneser–Freiman type for bounded-exponent subset sums, standing between Theorem 5.5 and its polylog tier.

5.4 Sharpness at the first link, and what lies past the ceiling

Two things remain to be said about the chain: that its first link is sharp, and that we know of exactly one route past the ceiling of Remark 5.6.

Sharpness comes from recognizing character failure as a *norm-form* condition in disguise. For $r \equiv 3 \pmod{4}$, quadratic reciprocity gives $(\frac{-r}{\ell}) = (\frac{\ell}{r})$ for odd primes $\ell \nmid r$, so

$$\text{every prime factor of } a \text{ has } \left(\frac{\cdot}{r}\right) = +1 \iff -r \text{ is a square mod } 4a \iff a \text{ is \textit{primitively} represented by a form } c \quad (5.1)$$

the last step by the correspondence between representations and square roots of the discriminant (Cox [16, Lem. 2.5]); for r composite the symbol is the Jacobi symbol throughout. For r prime the discriminant has a single genus and the whole form class group participates; at $r = 15$ there are two genera and two classes, and the count is applied per form and summed. The counting theorem is Iwaniec’s half-dimensional sieve [18] in prime-indexed form: the number of primes $p \leq N$ with $p = B\varphi(\xi, \eta) + A$ for a primitive form φ is $\asymp N/(\log N)^{3/2}$ [19], unobstructed by parity since $\beta(\frac{1}{2}) = 1$.

Theorem 5.9 (the first link is sharp). *For $R \in \{3, 7, 15\}$ the number of hard primes $p \leq x$ failing R is $\asymp x/(\log x)^{3/2}$, and the same holds with $\ll, \gg$ separately at $R = 11$. In particular*

the chain of Theorem 1.11 is sharp at its first link, and no exponent better than $3/2$ is available from any single residual: by Theorem 1.10 every failure-sufficient family has sieve dimension $\geq \frac{1}{2}$, so $3/2$ is the ceiling of the method, not an artifact of the argument.

Proof sketch. At these residuals the exact criteria make failure equivalent to an all-residue condition on the factors of $a = (p + R)/4$, which by (5.1) is primitive representability by a form of discriminant $-r$, $r = 3, 7, 15$. The upper bound is the Fundamental Lemma as in Theorem 1.11; the lower bound is [19] per class of forms, with (H1)–(H3) absorbed into A and B , and the passage from representability by *some* form to *primitive* representability is [17]. At $R = 11$ the budget failures of Theorem 1.5(b) are not of this shape, whence only the one-sided bounds. Sander [25] proved a lower bound of this order for a related insolubility count by the same sieve; the accessible record of that paper states no matching upper bound, and the upper bound here is Fundamental-Lemma routine. $\square$

Past the ceiling, the one route we know abandons fixed lists for a *p-adapted* one. Lemma 5.10 below holds at composite residuals as well, and read backwards it selects a residual at which the all-residue failure mode is impossible by construction: for a hard prime p with least Legendre non-residue $q = q(p)$ (necessarily $q \geq 11$ by Corollary 2.3), put $R_0 = (-p) \bmod 4q$, so that $R_0 \equiv 3 \pmod{4}$, $q \mid (p + R_0)/4$, and $(\frac{q}{R_0}) = (\frac{p}{q}) = -1$; the same holds at every rung of the ladder $R_j = R_0 + 4qj$.

Lemma 5.10 (composite reciprocity and Jacobi necessity). *Let $R \equiv 3 \pmod{4}$ be admissible and q an odd prime with $q \mid (p + R)/4$, $\gcd(q, R) = 1$. Then $(\frac{q}{R}) = (\frac{p}{q})$ for the Jacobi symbol. Consequently, if no prime factor of $a = (p + R)/4$ has $(\frac{\cdot}{R}) = -1$ then (1.2) fails at R .*

Proof. $q \mid p + R$ gives $R \equiv -p \pmod{q}$; Jacobi reciprocity with $R \equiv 3 \pmod{4}$ gives $(\frac{q}{R}) = (-1)^{\frac{q-1}{2}} (\frac{R}{q}) = (-1)^{\frac{q-1}{2}} (\frac{-1}{q}) (\frac{p}{q}) = (\frac{p}{q})$ — the proof of Theorem 1.9 verbatim, which never used primality of R . For the second claim, all factors of a Jacobi residues gives $(\frac{a}{R}) = +1$, and by consistency (1.3) also $(\frac{p}{R}) = +1$; so every divisor $k \mid m^2$ has $(\frac{k}{R}) = +1$ while the target has $(\frac{-m}{R}) = (\frac{-1}{R}) = -1$. $\square$

The construction performs well: over all 1,587,581 hard primes below 10^9 and window samples to 10^{11} , R_0 alone certifies 94.78% and 95.0–95.8% with no drift in p , and the ladder capped at $R_j \leq 400$ certifies 100% throughout; on the adversarial 10^{11} deep tail ($R_{\min} \geq 43$) the figures are 75.1% and 99.99%. Theorem 1.10 makes failure at each rung a sifting condition exactly as in Theorem 1.11, and Proposition 2.2 gives an unconditional first-rung decay $\ll_q (\log x)^{-\eta_q}$, $\eta_q > 1/(4q)$. Turning the measured per-rung decay into a theorem, however, needs the hypothesis that the true failure sets are a positive proportion of the necessary-condition sets containing them — and present methods bound that below only on a sparse family of classes (0.023% of first rungs below 2×10^7 ; when R_0 is prime, 65.2% of the time, Theorem 1.9 rules the mechanism out by construction). The gap is structural, not technical: budget failure is not a character event (at $R = 11$ the configurations with class set $\{2, 7\}$ and multiplicities $(1, 1)$ versus $(4, 2)$ agree under every Dirichlet character mod 11, yet the first misses the target and the second attains it, and 97.5% of measured first-rung failures are of that type), and weighted detection buys nothing, since by Proposition 2.2 the detector is already maximally relaxed and the first moment of any Bonferroni/Selberg weighting over the failure set is itself a failure count. Even granting the hypothesis, the payoff arrives only past Remark 5.6’s ceiling, at rung count $(\log x)^\theta$ with $\theta > 2/3$; at the rung count matching Theorem 5.5 it gives nothing Theorem 5.5 does not

already give unconditionally, with a better threshold. The full development and archives are in the accompanying repository.

6 Open problems

1. **Finite covering.** Prove that some finite admissible set S covers all hard primes. Theorem 1.13 gives density $(\log x)^{-A}$ for every A ; emptiness is beyond upper-bound sieving (Remark 5.4), and both the calibration and item 4 below predict that any fixed list eventually fails.
2. **Beyond fixed lists.** Theorem 5.5 carries list growth to length $\varepsilon \log \log x$, and what governs everything further is Conjecture 5.7: the aperiodic window-container statement, whose difficulty is localized to the control of non-critical deletion steps and whose resolution gives Theorem 5.8. The proof of Theorem 1.10 shows the extremal failing supports are *subgroup-trapped*; exploiting that structure uniformly in R — fewer, p -independent branches — is a well-posed additive-combinatorics question, and the structurally right target remains a slowly growing bound on $R_{\min}(p)$.
3. **The constant 5.17.** The joint $\{3, 7\}$ -exceptional density is measured as $\approx 5.17/\log x$, stable to 1% across three decades. Conjecturally $E(x) \sim Cx/(\log x)^2$ with $C = c_0 \prod_q \delta_q$, the local factors recording, for each prime q , the probability that q divides neither form in a forbidden class. Proving the asymptotic — not just the upper bound — needs a two-form Selberg–Delange or shifted-convolution analysis, and is a natural test problem for such methods.
4. **The record, and its growth.** Theorem 1.9 describes the record prime quantitatively (Corollary 2.3) and yields a conditional answer to item 1: under Dickson’s prime-tuple conjecture, for every B there are infinitely many hard primes failing every residual $\leq B$ — fix the forced smooth parts of the shifted values (character-neutral by Corollary 2.3(i)), make the cofactors prime in prescribed classes, and apply Proposition 2.1. So no fixed finite list suffices, conditionally, and the record grows without bound, at the pure-character level like $\log x / \log \log x$ — a rate the census of Section 4 is consistent with but far from resolving. Unconditional versions appear out of reach: already “infinitely many hard p fail $R = 3$ ” contains a two-form prime-tuple statement. What remains open is an unconditional upper bound on $R_{\min}(p)$ — and this is not incremental: by Proposition 1.1, $R_{\min}(p) \leq 2p$ is *equivalent* to the conjecture at p , so an unconditional bound $R_{\min}(p) \leq f(p)$ for any f whatsoever is the full problem.

A Verification details

This appendix documents every computer-assisted claim of the paper (their proof status is stated in Remark 1.8): the finite enumerations that are *proof components* (Appendix A.1), the large-scale certificate computations behind Section 4 and how each dataset is verified (Appendix A.2), and the commands that reproduce all of it (Appendix A.3). All code is in the accompanying repository; every check is deterministic and uses exact integer arithmetic only.

A.1 Proof components

Each component below is an exhaustive enumeration of a finite space whose sufficiency is established in the text (Theorem 1.3 for the exact criteria; for Lemma 5.2, monotonicity of the reachability of the target in support and multiplicities, checked at minimal exponents). Each run completes in seconds on commodity hardware, except the largest support-bound check ($R = 107$: 331 s); the entry points named are in the module `erdos_straus.theory`.

Trusted computing base. The certificate data does not rest on the solver: every certificate claim is re-checked by evaluating $4abc = p(bc + ca + ab)$ directly, so the trusted base for the data is a bignum multiply-and-compare, not the search pipeline. The finite case analyses trust more — the enumeration code itself, roughly sixty lines per check — and four mitigations apply. The enumerated spaces are defined by statements proved in the text (Theorem 1.3, Lemma 5.1), and the code mirrors those definitions one-to-one. Independent implementations agree where they overlap: the subset dynamic program and the direct binomial enumeration verify identical claims at $R = 19, 23$, and the exact criteria are cross-checked against ground-truth divisor search on up to 1,587,581 primes with zero discrepancies. Every check is deterministic, uses exact integer arithmetic only, and runs from the repository in seconds (331 s for the largest) — an independent re-implementation is an afternoon’s work, and we would welcome one.

Fourth, much of the development is formalized (`lean/`) in Lean 4 [3] with `mathlib` [21]. Every Lean proof is checked by a small trusted kernel, so a theorem it accepts is correct relative to its formal statement. Two mechanisms matter below: `native_decide` discharges a decidable proposition by running compiled code, extending the trusted base from the kernel to Lean’s compiler and evaluator; `#print axioms` reports exactly which axioms, including any such evaluator appeal, a proof uses. The formalization comprises:

- *The elementary layer*, as ordinary kernel-checked proofs: certificate soundness and integrality, both directions of Theorem 1.2, both families of Proposition 1.12, Theorem 1.9, and the ladder’s elementary layer — Lemma 5.10 in full, and the selected-residual identity of Section 5.4 (the formalization shows the coprimality hypothesis of Lemma 5.10 is not needed: for $q \mid R$ both symbols vanish).
- *The enumeration layer*: the reachability model underlying Theorem 1.3 is defined in the multiplicative structure of $\mathbb{Z}/R$ directly, and both halves of Lemma 5.1 are proved as theorems, with no computational content to trust. The finite checks behind Theorem 1.4 (1,536 configurations, checked as the 384 that remain after the neutral class collapses), Theorem 1.5 (497,664 capped configurations, including the exact two-case failure classification), and Lemma 5.2 at $R = 19, 23$ (437,580 resp. 7,759,752 checks), and Theorem 1.6 (349,920 configurations, via a product-index bridge for the non-cyclic unit group) are discharged by `native_decide`.
- *The bridges and the reduction*: the larger enumerations are *evaluated* in discrete-log coordinates for speed, but the translation to the multiplicative model is itself a proved theorem, so the coordinates add nothing to the trusted base. Separately, membership of the target class in the reachability model is proved equivalent to the existence of a divisor $k \mid m^2$ in that class, and produces the explicit certificate (k, b, c) solving (1.1) — closing the loop between the abstract model the enumerations check and the integer certificates the conjecture asks for.

- *Two capstones*: a composed corollary (if $(p + 19)/4$ has nine prime factors in pairwise-distinct unit classes other than 1 modulo 19, an explicit solution exists, machine-checked end to end), and Lemma 5.2 at $R = 31$ by *certified dynamic programming* — the $\binom{29}{15} = 77,558,760$ supports are never enumerated; a machine-checked induction shows the dynamic program covers every support, and the compiled evaluator runs the program and the final check (3,001 states). The program’s *correctness* is proved; its evaluation is what `native_decide` trusts.

Every main theorem is audited with `#print axioms`, and helper lemmas are covered transitively through them. The elementary layer is now formalized outright, including Proposition 1.1 in both directions (a solution is a certificate at its own residual, and conversely — so the equivalence organizing this paper is machine-checked) and Propositions 2.1 and 2.2. Theorem 1.6’s finite check is formalized by a product-index mask bridge over the non-cyclic $(\mathbb{Z}/15)^* \cong C_2 \times C_4$, and Theorem 1.10 fully symbolically: Kneser’s addition theorem vendored from its existing sorry-free formalization, an iterated many-summand form proved on top, and the support bound derived for an *arbitrary* finite abelian group with an explicit involution count — part (iii)’s general form, from which parts (i) and the odd-order variant follow as instances. So too the branch-classification layer: the single kernel step of §5.3 in all three parts, with the quotient *constructed* rather than assumed and the container size bound $\leq d/2 - 1$ derived from the support bounds downstairs; the puncture lemma in full; and the $2^{c\sqrt{d}}$ branch count end to end — the complete-partition machinery, the maximality of the constructed supports, their injectivity, and an explicit $2^{c\sqrt{d}}$ lower bound.

The transfer of Theorem 1.10(ii) out of the additive model is formalized as well, which makes Lemma 5.2 a machine-checked *corollary* of Theorem 1.10 rather than a separately verified statement. The chain is: membership in the budget-2 reach yields an explicit exponent decomposition with all exponents ≤ 2 ; a set of units of $\mathbb{Z}/R$ avoiding the identity carries at most one involution, since $x^2 = 1$ has only the roots ± 1 in a field, which collapses the general bound $\max(\lfloor (g + t - 2)/2 \rfloor, g/2 - 1)$ to $g/2 - 1$ at $G = (\mathbb{Z}/R)^*$ exactly as a single 2-torsion class does in the cyclic case; hence at least $(R - 1)/2$ factor classes other than the identity reach every unit class as a bounded-exponent product, which is realized by an honest integer divisor of a^2 (a product of bounded prime powers, dividing a^2 by pairwise coprimality); and composing with the divisor bridge gives, for *every* prime residual $R < p$ with $4a = p + R$, explicit positive $k \mid (pa)^2$, b, c satisfying $4abc = p(bc + ac + ab)$. The $R = 19$ composed corollary is the same conclusion at one residual, inheriting the `native_decide` axiom of the enumeration behind it; the general statement is symbolic throughout — no enumeration, no evaluator, no restriction on R . (Still not formalized: the identity $t = 2^{\omega(R)} - 1$ for the involution count at *composite* R , so the general bound is quoted there with that t supplied by hand; the *iterated* kernel reduction; and the analytic arguments of Sections 5.1 and 5.4, which rest on the hand proofs alone.) For these enumerations the sixty-line trust base above is thus replaced by the Lean toolchain evaluating a formally stated proposition — and the Python and Lean implementations, written independently against the same statements, agree. (The development notes and Lean sources use working letter names for several results; the repository’s documentation includes the dictionary.)

As sanity checks (not proof components), the criteria of Theorems 1.2–1.6 were compared against ground-truth divisor search on the complete solvability data below 10^9 : 1,587,581/1,587,581 cases at $R = 3$, 41,421/41,421 sampled failures at $R = 7$ (all character failures, as Theorem 1.4 requires), 158,759/158,759 at $R = 15$ and 79,380 at $R = 11$, with zero discrepancies; the general composite- R engine agrees on 7,938 sampled primes at every

component	configuration space	count	entry point
Thm. 1.4 ($R = 7$)	class multisets in $(\mathbb{Z}/7)^*$, multiplicities ≤ 3 , with the forced even factor, $p \bmod 7 \in \{1, 2, 4\}$, and the consistency relation	1,536	<code>verify_R7_finite</code>
Thm. 1.5 ($R = 11$)	dynamic program over $(\mathbb{Z}/11)^*$ with the forced factor 3; collapses to 25 equivalence states (16 succeed, 6 fail by character, 3 by budget)	25	<code>finite_criterion_dp</code>
Thm. 1.6 ($R = 15$)	class multisets in $(\mathbb{Z}/15)^*$, multiplicities ≤ 5 , with the forced even factor, $p \bmod 15 \in \{1, 4\}$, and the consistency relation (349,380 succeed, 540 fail all-residue, zero budget failures)	349,920	<code>verify_R15_finite</code>
Lem. 5.2 ($R = 19$)	all supports of size 9 over the nonzero logs $\times$ all classes of p , at minimal multiplicities	437,580	<code>verify_support_bound</code>
Lem. 5.2 ($R = 23$)	all supports of size 11, likewise	7,759,752	<code>verify_support_bound</code>
Lem. 5.2 ($R = 19$ –107, all twelve primes)	subset dynamic program over reachability-mask states; state counts 271 ($R=19$), 3,001 ($R=31$), 2,414,145 ($R=83$), 19,530,971 ($R=107$); zero violations throughout	$\approx 19.5\text{M}$	<code>verify_support_bound_dp</code>
Thm. 1.10(iii), $R \leq 107$ clause	maximal support of a proper product-set in $(\mathbb{Z}/R)^*$ equals $\varphi(R)/2 - 1$ for every composite admissible $R \leq 107$ (e.g. $R = 95$: max 35 over 583,455 realizable product-sets)	12 moduli	<code>kneser_support_general</code>

Table 5: Machine-verified proof components: enumerated configuration spaces and the entry points in `erdos_straus.theory`.

composite admissible residual; Theorem 1.9 and Corollary 2.3 were checked on 135,588 resp. 61,438 instances with zero violations. By contrast `kneser_support_general` is *not* a sanity check but a proof component of the $R \leq 107$ clause of Theorem 1.10(iii), listed as such in Table 5.

A.2 Certificate datasets

The datasets underlying the census, and the verification applied to each, are tabulated in Section 4.1 (Table 2); all are archived in the repository under `data/`, with sha256 manifests.

A.3 Reproduction

With the repository installed (`pip install -e .`), the commands of Table 6 regenerate and re-verify everything above. Generation of the full 10^{10} map takes about 11 minutes on four cores; every other command runs in seconds to minutes.

claim	entry point
Thms. 1.4, 1.6 finite checks	<code>verify_R7_finite()</code> , <code>verify_R15_finite()</code>
Thm. 1.5 enumeration	<code>finite_criterion_dp(11, forced_logs=[...])</code> (as in <code>tests/test_solver.py</code>)
Lem. 5.2 at $R = 19, 23$	<code>verify_support_bound(19), (23)</code> ; larger R via <code>verify_support_bound_dp</code>
Thm. 1.10 confirmations	<code>verify_support_bound_strong(31)</code> , <code>kneser_support_general(15)</code>
generate the 10^{10} map	<code>python -m erdos_straus.bulk_generate -max 10000000000 -segmented -store rmap</code> (≈ 11 min on four cores)
regenerate the 10^{12} census	<code>scripts/run_1e12.sh</code> (see <code>scripts/RUN_1E12.md</code>)
verify a dataset	<code>python -m erdos_straus.verify data/<file></code>
distribution / covering / taxonomy	<code>python -m erdos_straus.analyze dist cover tail -rmap data/hard_primes_1e9_minimalR.json.gz;</code> <code>python -m erdos_straus.theory</code>
selected-residual census	<code>python -m erdos_straus.burgess_scan</code> (archives: <code>data/analysis/burgess_*.json</code>)
Lean formalization	<code>cd lean/ErdosStraus && lake exe cache get && lake build</code>

Table 6: Reproduction entry points; the unqualified names are Python functions in the module `erdos_straus.theory`.

The full test suite (`python -m pytest`; 63 tests) exercises the proof-component checks, the exact criteria against ground truth, and sampled dataset validation on every run. Table 7 summarizes the computational environment.

implementation	Python 3, exact integer arithmetic only; no external libraries in any proof-component check. Factorization by trial division against numpy-sieved prime tables
wall-clock	10^{12} generation: 10.5 h on 32 cores (AMD Ryzen AI Max+ 395), of which 53 min sieving; 10^{11} : 2.85 h on 4 cores; 10^{10} : 656 s; 10^9 : 49 s; largest proof-component check ($R = 107$ dynamic program): 331 s
verification	every certificate re-checked via $4abc = p(bc+ca+ab)$; exhaustive below 10^9 ; 2,350,431 samples plus all 118,210 tail entries at 10^{12} (Table 2)
integrity	datasets archived in the repository; prime arrays and R -sequences pinned by SHA-256 in the corresponding <code>meta.json</code> (at 10^{12} , prefixes <code>ae1e4e76...</code> , <code>7736d10b...</code>); the 304 MB 10^{12} value array is regenerated deterministically by <code>scripts/run_1e12.sh</code> and checked against its stored digest
formal layer	Lean 4.32.2 + mathlib (pinned toolchain); six <code>native_decide</code> checks; Theorem 1.10(i) fully symbolic (vendored Kneseer); every main theorem audited by <code>#print axioms</code>

Table 7: Computational environment and integrity summary.

Data and code availability

All code, all datasets of Table 2, and the Lean development are openly available in the repository <https://github.com/jeffmhopkins/Erd-s-Straus-attack>.

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